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Dynamics of rational billiards

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Introduction

Motivation and background

Billiards are remarkably rich instances of dynamical systems, which are systems characterized by how they evolve, particularly for large amounts of time. Plainly speaking, we have a billiard table of any shape and we want to know how the ball will behave after shooting it in a direction. Of course, saying *any* shape is too broad for mathematicians, so we will specify what is allowed in Chapter 2. We can see four examples of billiard systems in Figure 1.

Interestingly, the dynamical properties of billiards depend on the curvature of its sides. In fact, the techniques used to analyze its trajectories are radically different in each case. For instance, in convex billiards a prominent role is played by the KAM theory and the theory of area preserving twist maps. Nevertheless, rational polygons have been the most intensely studied since they bring together geometric topology, dynamical systems, algebraic geometry and much more.

Suppose we shoot the ball. Since the (specular) law of reflection is well-known, the movement is deterministic. However, as usually happens with dynamical systems, it becomes barely tractable as a long time passes. Within this informal setting there may arise some primitive questions about billiards:

Q1. In an arbitrary billiard, is there always a periodic trajectory?

Q2. If a trajectory is not periodic, is it dense? That is, is there a dichotomy between these two properties?

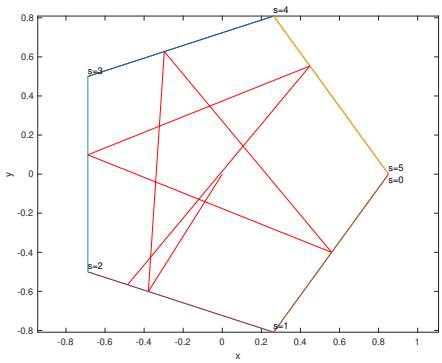
Q3. What is the typical behavior of a trajectory?

Q4. Are trajectories typically equidistributed?

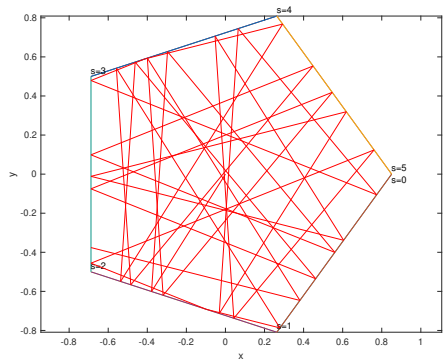
Q5. What is the structure of periodic trajectories and how does it relate to the geometry of the boundary of the billiard?

Q6. How do perturbations of the boundary affect the structure of the trajectories?

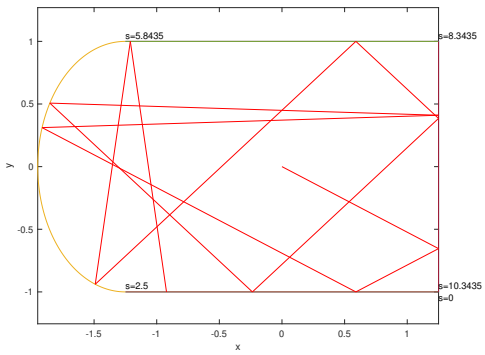
Q7. How can one characterize the complexity of the trajectory structure?



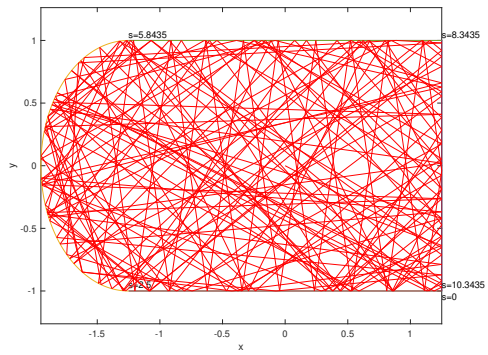
(a) Billiard in a pentagon with 5 collisions.



(b) Billiard in a pentagon with 25 collisions.



(c) Billiard in a half stadium with 10 collisions.



(d) Billiard in a half stadium with 150 collisions.

Figure 1: Billiard systems generated with the software *Billiard Simulator* of Mason Porter. Retrieved July 21, 2025 from https://www.mathworks.com/matlabcentral/fileexchange/10692-billiard-simulator.

Therefore, we are *a priori* interested in the qualitative properties of the asymptotic behaviour of trajectories in terms of topology and measure theory. Actually, the answers to all of the questions above are unknown for general billiards, although much is known about rational ones. Concretely, with our exposition we will be able to answer Q3 and Q4.

Historically, the subject has been long studied since the seminal work of Poincaré [Poi05] and Birkhoff [Bir27] (see also [Had98],[Mor21] and [Hed39]) and later reassumed by Schwartzman [Sch57]. Zemlyakov and Katok [ZK75] were the first authors to treat rational billiards. They introduced the unfolding that now carries their name (which some authors attribute to Fox and Kershner [FK36]) and which was a breakthrough without which none of the subsequent results would have been possible. Apart from this, they proved the topological transitivity of billiard trajectories. Afterwards, Boldrighini, Keane and Marchetti [BKM78] strengthened the previous result to explain that the typical be-

haviour of a billiard trajectory is to become dense in the table as time goes to infinity. These aforementioned papers remain the most influential in the field.

The first strong measure-theoretic results appeared with the work of Masur [HM79; Mas82; Mas85] and Veech [Vee78; Vee86; Vee89], among others, on Teichmüller theory, more concretely on quadratic differentials, measured foliations and interval exchange transformations. Kerfchoff, Masur and Smillie [KMS86] proved that the vertical trajectories of quadratic differentials are uniquely ergodic for almost every initial direction, and thus the billiard flow is uniquely ergodic for almost every initial direction. The same year Masur [Mas86] proved that every rational polygon contains at least one periodic trajectory. In the same line Boshernitzan, Galperin, Krüger and Troubetzkoy [Bos+98] showed that not only periodic trajectories exist, but periodic points are dense in the phase space of the billiard flow. More advanced results connecting billiards with moduli spaces and algebraic geometry have been led by Eskin, McMullen and Mirzakhani (see, for instance, [FT02], [McM03] or [EM18]); however, they are far off the humble pretensions of this text.

There are also excellent surveys such as [Gut86], [Smi00], [De 11] or [Sch23], as well as introductory texts: [Tab91], [Kat05], [Wri14], [Gar20], with special mention to the monograph of Chernov and Markarian [Che14]. This text, in spite of dealing with chaotic billiards, provides a unified and formal treatment of the basic objects such as the phase space, the billiard flow and Poincaré first return map. Finally, I want to mention the work of Forni and Matheus [FM15], Farb and Margalit [Far12] and Strebel [Str84], for their pedagogical impact and presence throughout this text, specially in understanding the nature of quadratic differentials. At the same time, gratitude is in order for the work of Yoccoz [Yoc10], Zorich [Zor06] and Massart [Mas22], in this case for the many revealing explanations about flat structures.

Research goals and overview

The research goals of this thesis are, as was mentioned, humble. We want to reproduce essentially two results regarding rational billiards. Firstly, we show the billiard flow is minimal in almost every initial direction by using interval exchange transformations. This means that from every initial point, almost every direction gives rise to a trajectory that is dense in the table, i.e., the trajectory gets arbitrarily close to every point. Secondly, we prove the billiard flow is uniquely ergodic. The later result implies the former, but we have decided to keep the first half of the exposition (Chapter 1 and Chapter 2) as elementary as possible, whereas the second half (Chapter 3 and Chapter 4) carries all the heavy tools.

It should not be overlooked that a significant part of the motivation behind this research is the pursuit of deeper self-understanding and, hopefully, that of the reader, of the intricate connections between billiards, flat structures, quadratic differentials, and measured foliations. The value of this work relies not on original results nor new ideas in the field, but on the gathering of sparse information and the explanation of ideas that sometimes are considered too elementary to be covered in surveys. Even so, we give original proofs of the following results:

- Proposition 2.6, in which the non-trivial observations of §2.1.2 play a crucial role.
- Theorem 2.12, which in turn implies the billiard flow is not minimal is the phase space.
- Proposition 2.14, which has the novelty of avoiding the Zemlyakov-Katok unfolding.

While these results are not discoveries, their worth lies in the fact that they are taken for granted in all the literature, with the exception of Proposition 2.14, which is proved by other techniques. Furthermore, with the exception of Figure 3.2, all the figures are original.

The narrative will be deductive, from the general to the particular. Two details are to be highlighted. First, the exposition has been tailored so that in §4.1, the translation structure arising from the Zemliakov-Katok unfolding emerges as an evident consequence of the construction. Second, it is worth mentioning that we have purposefully postponed introducing ZK unfolding until Chapter 4 since there was no need of extra structure, so minimality is shown by interval exchange transformations alone.

In the Chapter 1 we set up the background about dynamical systems and orbits. Since ergodicity and unique ergodicity depend upon measure preserving maps, we prove in Corollary 1.15 that any continuous transformation of a compact metric space admits an invariant Borel probability measure. Chapter 2 serves a closing of the first conceptual block of the thesis: we present rational billiards and close the minimality problem. The heart of the text is Chapter 3. This chapter is essentially descriptive, that is why it does not contain any explicit theorem-proof environment and some parts should be read like a survey. In it we mention, for the sake of their existence, the moduli space of Riemann surfaces and the Teichmüller space. Afterwards, quadratic differentials are thoroughly explained: the (almost) flat metric they induce and the action of $\mathrm{SL}(2,\mathbb{R})$ which deforms this metric. One key element to understand quadratic differentials is to think geometrically and compute analytically, as we do with linear maps between vector spaces. Ultimately, in Chapter 4 we link all the apparently useless objects presented up to the moment to rational billiards by means of the Zemlyakov-Katok unfolding, and finally prove the main result. More is to be found on the header of each chapter.

Symbols and Notation

$O(2)$	Orthogonal group
$\mathrm{SO}(2)$	Group of rotations of $\mathbb{R}^2$
$\mathrm{SL}(2,\mathbb{R})$	Group of real matrices of determinant 1
$\mathring{X}$	Topological interior
$\overline{X}$	Topological closure
π_1	Projection onto the first factor of a cartesian product
χ	Euler characteristic
g	Genus
$G \curvearrowright X$	Action of the group G on X
$\mathcal{P}$	Billiard table
$\partial\mathcal{P}$, Γ	Boundary of $\mathcal{P}$
Γ_*	Vertices of $\mathcal{P}$
$\tilde{\Gamma}$	Regular boundary points of $\mathcal{P}$
$d(\cdot,\cdot)$	Euclidean metric
q	Quadratic differential
$\mathcal{B}(X)$	Borel σ-algebra of X
$M(X)$	Set of all Borel probability measures on X
$M(X,T)$	Borel probability measures for which T is measure-preserving
Ω	Phase space for the billiard flow
$\tilde{\Omega}$	Points of Ω for which the billiard flow is defined for all $t \in \mathbb{R}$
ζ	Natural parameter for a quadratic differential
G	Group of linear reflections across the sides of $\mathcal{P}$
$H \trianglelefteq G$	H is a normal subgroup of G
$\langle\cdot,\cdot\rangle$	Euclidean inner product
S^1	Unit circle
Mod_g	Mapping class group for surfaces of genus g

Chapter 1

Preliminaries

In this chapter we provide the appropriate background and tools to study billiard dynamics. First, we make an introduction to measure theory and measure-preserving transformations, where the main results are the Markov-Kakutani theorem (Theorem 1.14) and the Bogoliubov-Krilov theorem (Corollary 1.15). The principal references for this section are [RF10] and [Wal00].

Afterwards, we fix notation about dynamical systems, present the basic notions of topological dynamics and ergodic theory and some interesting connections between both fields. The heart of matter are the concepts of minimality (Definition 1.20), topological transitivity (Definition 1.24), ergodicity (Definition 1.26) and unique ergodicity (Definition 1.30). The main reference here is [Wal00].

1.1 Measure theory

Definition 1.1. A *measurable space* is a pair $(X, \mathcal{M})$ consisting of a set X and a σ-algebra $\mathcal{M}$ of subsets of X. A subset E of X is called measurable provided E belongs to $\mathcal{M}$.

Definition 1.2. A *measure* μ on a measurable space $(X, \mathcal{M})$ is an extended real-valued nonnegative set function $\mu : \mathcal{M} \to [0, \infty]$ for which $\mu(\varnothing) = 0$ and which is countably additive in the sense that for any countable disjoint collection $(E_k)_{k\geq 1}$ of measurable sets,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

By a *probability space* $(X, \mathcal{M}, \mu)$ we mean a measurable space $(X, \mathcal{M})$ together with a measure μ defined on $\mathcal{M}$ such that $\mu(X) = 1$. The measure μ is called a *probability measure*.

Definition 1.3. For a measurable space $(X, \mathcal{M})$, a mapping $T : X \to X$ is said to be a *measurable transformation* provided for each measurable set E, $T^{-1}E$ is also measurable.

Definition 1.4. For a measure space $(X, \mathcal{M}, \mu)$, a measurable transformation $T : X \to X$ is said to be *measure-preserving* provided $\mu(T^{-1}E) = \mu(E)$, for all $E \in \mathcal{M}$. We say that T preserves μ or that μ is T*-invariant*.

Let X be a compact metrisable space and $\mathcal{B}(X)$ its Borel σ-algebra, i.e., the smallest σ-algebra that contains all the open sets of X. We shall denote by $M(X)$ the collection of all Borel probability measures defined on $(X, \mathcal{B}(X))$, which is convex and non-empty, since each $x \in X$ determines a Dirac measure $\delta_x \in M(X)$ defined by

$$\delta_x(A) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A. \end{cases} \tag{1.1}$$

It is a standard fact that a Borel probability measure μ on a metric space X is regular (see [RF10, Theorem 20, §21.6]), i.e, for every $B \in \mathcal{B}(\mathrm{X})$ and for each $\epsilon > 0$ there exists an open set U_ϵ and a closed set C_ϵ with $C_\epsilon \subseteq B \subseteq U_\epsilon$ such that $\mu(U_\epsilon \setminus C_\epsilon) < \epsilon$. As a consequence (see [Wal00, Corollary 6.1.1]), for any $B \in \mathcal{B}(X)$

$$\mu(B) = \sup_{C \subseteq B \text{ closed}} \mu(C). \tag{1.2}$$

We denote by $C(X)$ and $C(X)^*$ the space of continuous functions $X \to \mathbb{R}$ and its dual, respectively.

Proposition 1.5 ([Wal00, Theorem 6.2])**.** *Let X be a compact metric space and $m, \mu \in M(X)$. If $\int_X f\,dm = \int_X f\,d\mu$ for every $f \in C(X)$, then $m = \mu$.*

Proof. Because of Equation (1.2), it suffices to show that $m(C) = \mu(C)$ for every closed set $C \subseteq X$. Suppose C is closed and let $\epsilon > 0$. Again, by the regularity of m there exists an open set U with $C \subset U$ and $m(U \setminus C) < \epsilon$.

Define $f : X \to \mathbb{R}$ by

$$f(x) = \begin{cases} 0 & \text{if } x \notin U \\ \dfrac{d(x, X \setminus U)}{d(x, X \setminus U) + d(x, C)} & \text{if } x \in U. \end{cases}$$

This is well-defined since the denominator is not zero. Also f is continuous, $f = 0$ on $X \setminus U$, $f = 1$ on C, and $0 \leq f(x) \leq 1$ anywhere else on X. Hence

$$\mu(C) \leq \int_X f \mathrm{d}\mu = \int_X f \mathrm{d}m \leq m(U) + \epsilon.$$

Therefore, $\mu(C) < m(C) + \epsilon$, for every $\epsilon > 0$, so $\mu(C) \leq m(C)$. By symmetry we have that $m(C) \leq \mu(C)$. □

Definition 1.6. Let X be a compact metric space. A map $\Lambda : C(X) \to \mathbb{R}$ is said to be a *normalised nonnegative linear functional* if

(i) $\Lambda(1) = 1$, $\Lambda(f) \geq 0$ whenever $f \geq 0$,

(ii) $\Lambda(af + bg) = a\Lambda(f) + b\Lambda(g)$ for all, $a, b \in \mathbb{C}$ and $f, g \in C(X)$.

Theorem 1.7 (Riesz-Markov, [Par05, Corollary 4.6.5]; cf. [RF10, p. 458])**.** *Let X be a compact metric space. To every normalised nonnegative linear functional Λ on $C(X)$ there exists $\mu \in M(X)$ such that*

$$\Lambda(f) = \int_X f\, d\mu, \quad \forall f \in C(X).$$

We do not include a proof of Theorem 1.7 since it is technically involved and would take us too far afield from the main narrative.

Let X be a compact metric space. Denote by $\Omega(X)$ the collection of all normalised positive linear functionals on $C(X)$. Then, there is a bijection

$$\begin{aligned} \Theta : M(X) &\longrightarrow \Omega(X) \subset C(X)^* \\ \mu &\longmapsto \Lambda_\mu. \end{aligned}$$

Injectivity follows from Proposition 1.5 and surjectivity from Theorem 1.7. Clearly, Θ is affine and

$$\|\Lambda_\mu\|_{C(X)^*} = \sup_{\|f\|=1} |\Lambda_\mu(f)| = \sup_{\|f\|=1} \left| \int_X f \mathrm{d}\mu \right| \leq \sup_{\|f\|=1} \int_X |f| \mathrm{d}\mu = 1, \quad \text{for } \Lambda_\mu \in \Omega(X),$$

so $M(X)$ is identified with a convex subset of the unit ball in $C(X)^*$. This allows to inherit a topology on $M(X)$ from the weak-* topology on $C(X)^*$ (see [RF10, p. 275]) which is independent of any metric chosen on X. This topology on $M(X)$ is the smallest one making the maps $\mu \mapsto \int_X f \mathrm{d}\mu$ continuous, for $f \in C(X)$. A basis is given by all sets of the form

$$V_\mu(f_1, \ldots, f_k; \epsilon) = \left\{ m \in M(X) : \left| \int_X f_i \mathrm{d}m - \int_X f_i \mathrm{d}\mu \right| < \epsilon,\ 1 \leq i \leq k \right\},$$

where $\mu \in M(X)$, $k \geq 1$, $f_i \in C(X)$ and $\epsilon > 0$.

Proposition 1.8 ([Wal00, Theorem 6.4])**.** *If X is a compact metrisable space then the space $M(X)$ is metrisable in the weak-* topology. If $(f_n)_{n\geq 1}$ is a dense subset of $C(X)$ then*

$$D(m, \mu) = \sum_{n=1}^{\infty} \frac{\left| \int_X f_n\, dm - \int_X f_n\, d\mu \right|}{2^n \|f_n\|}$$

is a metric on $M(X)$ giving the weak- topology.*

Proof. It is easy to check that $D : M(X) \times M(X) \to \mathbb{R}$ is a metric. Fix i and consider the map $\mu \mapsto \int_X f_i \mathrm{d}\mu$, which is continuous because $\left| \int_X f_i \mathrm{d}m - \int_X f_i \mathrm{d}\mu \right| \leq 2^i \|f_i\| D(m, \mu)$. Since $(f_n)_{n\geq 1}$ is dense in $C(X)$ it follows that for each $f \in C(X)$ the map $\mu \mapsto \int_X f \mathrm{d}\mu$ is continuous on $(M(X), D)$. Therefore, we have proven that every open set in the weak-*

topology is open in $(M(X), D)$. To see the other inclusion consider $\mu \in M(X)$ and $\epsilon > 0$. Choose N so that $\sum_{n=N+1}^{\infty} \frac{2}{2^n} < \frac{\epsilon}{2}$ and let

$$\delta = \frac{\epsilon}{2} \left(\sum_{n=1}^{N} \frac{1}{2^n \|f_n\|} \right)^{-1}.$$

Then, $V_\mu(f_1, \ldots, f_N; \delta) \subset B_D(\mu; \epsilon)$. □

The following proposition is a direct consequence of Banach-Alaoglu's theorem (see [RF10, p. 299]). However, for the sake of keeping the preliminaries minimal we give a direct proof.

Proposition 1.9 ([Wal00, Theorem 6.5])**.** *If X is a compact metrisable space then $M(X)$ is compact in the weak-* topology.*

Proof. We write $\mu(f) = \int_X f \mathrm{d}\mu$. Let $(\mu_n)_{n\geq 1}$ be a sequence in $M(X)$. We show it has a convergent subsequence. We will drop the $n \geq 1$ subindex for the sequences throughout the rest of the proof, in favor of notational lightness.

Choose (f_n) dense in $C(X)$. Consider the sequence of complex numbers given by $(\mu_n(f_1))$. This is bounded by $\|f_1\|$, and so has a convergent subsequence, say $(\mu_n^{(1)}(f_1))$. Consider the sequence of numbers $(\mu_n^{(1)}(f_2))$; this is bounded and so has a convergent subsequence $(\mu_n^{(2)}(f_2))$. Notice that $(\mu_n^{(2)}(f_1))$ also converges. We proceed in this manner, and for each $i \geq 1$, construct a subsequence $(\mu_n^{(i)}(f))$ of (μ_n) such that

$$(\mu_n^{(i)}) \subseteq (\mu_n^{(i-1)}) \subseteq \cdots \subseteq (\mu_n^{(1)}) \subseteq (\mu_n),$$

and so that $(\mu_n^{(i)}(f))$ converges for $f = f_1, f_2, \ldots, f_i$. Consider the diagonal $(\mu_n^{(n)})$. The sequence $(\mu_n^{(n)}(f_i))$ converges for all i; thus, $(\mu_n^{(n)}(f))$ converges for all $f \in C(X)$ (by an easy aproximation argument). Let $J(f) = \lim_{n\to\infty} \mu_n^{(n)}(f)$. Clearly $J : C(X) \to \mathbb{C}$ is linear and bounded, as $|J(f)| \leq \|f\|$. Also $J(1) = 1$, and if $f \geq 0$, then $J(f) \geq 0$. By Riesz-Markov theorem (Theorem 1.7), there exists $\mu \in M(X)$ such that $J(f) = \int_X f \mathrm{d}\mu$ for all $f \in C(X)$, i.e.,

$$\int_X f \mathrm{d}\mu_n^{(n)} \longrightarrow \int_X f \mathrm{d}\mu.$$

□

Let $T : X \to X$ be a continuous transformation of the compact metrisable space X. We consider the pushforward $T_* : M(X) \to M(X)$ given by $(T_*\mu)(B) = \mu(T^{-1}B)$.

Lemma 1.10 ([Wal00, Lemma 6.6])**.** *Let X be a compact metrisable space and $\mu \in M(X)$. For every $f \in C(X)$, $\int_X f \, d(T_*\mu) = \int_X f \circ T \, d\mu$.*

Proof. It suffices to deal with real-valued $f \in C(X)$. By definition of T_* we have

$$\int_X \chi_B \mathrm{d}(T_*\mu) = (T_*\mu)(B) = \mu(T^{-1}B) = \int_X \chi_B \circ T \mathrm{d}\mu.$$

From this, it holds for simple functions, then for non-negative measurable functions by choosing an increasing sequence of simple functions converging pointwise. Finally, the formula holds for any continuous function $f : X \to \mathbb{R}$ considering the positive and negative parts of f. □

Proposition 1.11 ([Wal00, Theorem 6.7])**.** *The map $T_* : M(X) \to M(X)$ is continuous and affine.*

Proof. If $f \in C(X)$ by Lemma 1.10 we have $\int_X f \mathrm{d}(T_*\mu) = \int_X f \circ T \mathrm{d}\mu$. Therefore, if $\mu_n \to \mu$ in $M(X)$, then $\int_X f \mathrm{d}(T_*\mu_n) = \int_X f \circ T \mathrm{d}\mu_n \to \int_X f \circ T \mathrm{d}\mu = \int_X f \mathrm{d}(T_*\mu)$, and so $T_*\mu_n \to T_*\mu$. (see [Wal00, p. 149]).

If $m, \mu \in M(X)$ and $p \in [0,1]$ then $T_*(pm + (1-p)\mu)(B) = pm(T^{-1}B) + (1-p)\mu(T^{-1}B) = (pT_*m + (1-p)T_*\mu)(B)$, for every $B \in \mathcal{B}(X)$. □

Let $M(X,T) = \{\mu \in M(X) : T_*\mu = \mu\}$. This is the set of all Borel probability measures that make T a measure-preserving transformation of $(X, \mathcal{B}(X), \mu)$. We want to see that $M(X,T)$ is non-empty.

Proposition 1.12 ([Wal00, Theorem 6.8])**.** *Let X be a compact metrisable space. If $T : X \to X$ is continuous and $\mu \in M(X)$ then $\mu \in M(X,T)$ if and only if $\int_X f \circ T d\mu = \int_X f d\mu$, for every $f \in C(X)$.*

Proof. Follows from Lemma 1.10 and Proposition 1.5. □

Definition 1.13. A *locally convex space* is a topological vector space whose topology is defined by a family of seminorms P such that $\bigcap_{p \in P}\{x : p(x) = 0\} = (0)$, that is, the zero vector.

Note that the second condition forces a locally convex space to be Hausdorff. In fact, suppose that $x \neq y$. Then there is $p \in P$ such that $p(x-y) \neq 0$. Let $p(x-y) > \epsilon > 0$. If $U = \left\{z : p(x-z) < \frac{1}{2}\epsilon\right\}$ and $V = \left\{z : p(y-z) < \frac{1}{2}\epsilon\right\}$, then $U \cap V = \varnothing$ and U and V are neighborhoods of x and y, respectively.

Theorem 1.14 (Markov-Kakutani, cf. [Con85, §V.10])**.** *If K is a non-empty convex compact subset of a locally convex space and $F : K \to K$ is a continuous affine transformation, then there is an $x_0 \in K$ such that $F(x_0) = x_0$.*

The proof is omitted for space reasons but follows standard techniques. Since $C(X)^*$ is a locally convex space we have the following important result.

Corollary 1.15 (Bogoliubov-Krilov, see [RF10, Theorem 12, §22.4])**.** *If $T : X \to X$ is a continuous transformation of a compact metric space X, then $M(X,T)$ is non-empty.*

1.2 Dynamical systems

Dynamical systems deal with qualitative and quantitative asymptotic behaviour of deterministic systems, which are usually highly sensitive to initial conditions. Here we make these ideas precise. The primary reference for the general theory is [KH95], although some elementary definitions are from [BS05].

Definition 1.16 (cf. [GM12])**.** A *dynamical system on* L is a pair $(\mathcal{M}, \{\phi^t\}_{t\in T})$ such that

(i) $L = (T, +)$ is a totally ordered semigroup, where T is its *time set* and L its *time model*. Any $t \in T$ is called the *duration* of the system.

(ii) $\mathcal{M}$ is a non-empty set. Any $x \in \mathcal{M}$ is called a *state* of the system, and $\mathcal{M}$ is called its *phase space*.

(iii) $\{\phi^t : \mathcal{M} \to \mathcal{M}\}_{t\in T}$ is a family of maps indexed by T. For any $t \in T$, the map ϕ^t is called the *evolution law* of duration t of the system.

(iv) For any $v, t \in T$ and for any $x \in \mathcal{M}$

a) $\phi^0(x) = x$,

b) $\phi^{v+t}(x) = \phi^v(\phi^t(x))$.

Definition 1.17. A *flow* is a dynamical system $(\mathcal{M}, \{\phi^t\}_{t\in\mathbb{R}})$ on $(\mathbb{R},+)$ such that the one-parameter family of transformations $\{\phi^t : \mathcal{M} \to \mathcal{M}\}_{t\in\mathbb{R}}$ defines an action

$$\begin{aligned} \Phi : \mathcal{M} \times \mathbb{R} &\longrightarrow \mathcal{M} \\ (x, t) &\longmapsto \phi^t(x). \end{aligned} \tag{1.3}$$

Remark 1.1. The notation for a generic dynamical system $(\mathcal{M}, \{\phi^t\}_{t\in T})$ on $(T, +)$ may be abbreviated as (M, Φ) on T or $(M, \Phi)_T$ by analogy with (1.3).

Definition 1.18. Let $x \in \mathcal{M}$. The *forward orbit* of x is the set $\mathcal{O}^+_\phi(x) = \bigcup_{t\geq 0} \phi^t(x)$. If the flow is reversible, the *backward orbit* of x is defined as $\mathcal{O}^-_\phi(x) = \bigcup_{t\leq 0} \phi^t(x)$. The *orbit* of x is then given by

$$\mathcal{O}_\phi(x) = \mathcal{O}^+_\phi(x) \cup \mathcal{O}^-_\phi(x) = \bigcup_{t\in T} \phi^t(x).$$

Definition 1.19. A point $x \in \mathcal{M}$ is called a *periodic point* with minimal period $k > 0$ if $\phi^k(x) = x$ and, whenever $\phi^s(x) = x$ for some $s > 0$, it follows that $k \leq s$. The orbit of a periodic point is called a *periodic orbit*. A point $x \in \mathcal{M}$ is called a *fixed point* if it remains unchanged under the flow, that is, $\phi^t(x) = x$ for all $t \in T$.

There is a special way in which one can obtain a map from a flow. Let $\Phi : \mathcal{M} \times \mathbb{R} \to \mathcal{M}$ be a flow and consider $A \subset \mathcal{M}$ such that for every $x \in \mathcal{M}$ the set

$$T_x = \{t \in \mathbb{R}^+ : \phi^t(x) \in A\}$$

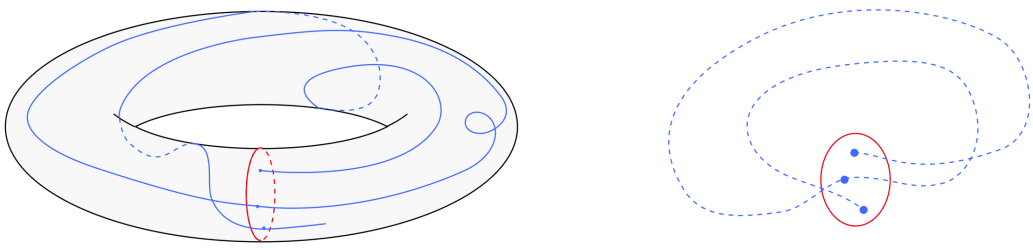


Figure 1.1: Flow on a toral phase space and the Poincaré map for a cross-section.

is non-empty and discrete. For $a \in A$, let $\tau(a) = \min T_a$, i.e., the first return time to A. Then, the *first return map* or *Poincaré map* $g : A \to A$ is defined by

$$g(a) = \phi^{\tau(a)}(a). \tag{1.4}$$

Intuitively, A is a cross-section of the phase space and by this procedure, we are restricting the flow on the whole $\mathcal{M}$ to a map on A (see Figure 1.1).

Remark 1.2. In virtually every interesting situation the phase space $\mathcal{M}$ possesses certain structure which is preserved by the evolution law. Katok and Hasselblatt highlight in [KH95] the main branches arising from the dynamics of different structure-preserving transformations, e.g. topological dynamics, ergodic theory (measurable dynamics), symplectic dynamics or differentiable dynamics.

In categorical terms, the evolution law is an endomorphism of the phase space (see [Tao08]). Thus, in the spirit of [GM12], Definition 1.16 would adjust to a *$\mathcal{C}$-dynamical system on L*, where $\mathcal{C}$ is a concrete category. Although this notation is more compact, we will specify the structure of each phase space and assume subsequently that the evolution law is structure-preserving.

1.2.1 Topological dynamics

Let X be a compact metric space and $T : X \to X$ a homeomorphism. Consider the dynamical system (X, T) on $\mathbb{Z}$.

Definition 1.20. T is called *minimal* if $\overline{\mathcal{O}_T(x)} = X$, for every $x \in X$.

A subset E of X is called *T-invariant* if $TE = E$.

Proposition 1.21 ([Wal00, Theorem 5.1])**.** *The following are equivalent:*

(i) T is minimal.

(ii) The only closed T-invariant subsets E of X are $\varnothing$ and X.

(iii) For every non-empty open subset U of X we have $\bigcup_{-\infty}^{\infty} T^n U = X$.

Proof. (i) $\Rightarrow$ (ii) Suppose T is minimal and let E be closed, non-empty and T-invariant. If $x \in E$ then $\mathcal{O}_T(x) \subset E$, and from the monotonicity of the closure we have that $X = \overline{\mathcal{O}_T(x)} \subseteq E$. Therefore, $E = X$.

(ii) $\Rightarrow$ (iii) If U is non-empty and open then $E = X \setminus \bigcup_{-\infty}^{\infty} T^n U$ is closed and $TE = T\left(X \setminus \bigcup_{-\infty}^{\infty} T^n U\right) = X \setminus \bigcup_{-\infty}^{\infty} T^n U = E$. Since $E \neq X$ we have $E = \varnothing$.

(iii) $\Rightarrow$ (i) Let $x \in X$ and let U be a non-empty open subset of X. Then, $\bigcup_{-\infty}^{\infty} T^n U = X$ and there exists $n \in \mathbb{Z}$ such that $x \in T^n U$. Hence, $T^{-n}x \in U$ and $\mathcal{O}_T(x)$ is dense in X.

□

Definition 1.22. A closed T-invariant subset E of X is called a *minimal set* with respect to T if $T|_E$ is minimal.

Zorn's lemma: *If any totally ordered subset of a partially ordered set $(S, \leq)$ has a lower bound in S, then S has a minimal element m. That is, $m \leq s$ for every $s \in S$ that is comparable to m.*

Proposition 1.23 ([Wal00, Theorem 5.2])**.** *T has a minimal set.*

Proof. Let $\mathcal{S}$ denote the collection of all closed, non-empty T-invariant subsets of X. Clearly $\mathcal{S} \neq \varnothing$, since $X \in \mathcal{S}$. The pair $(\mathcal{S}, \subseteq)$ is a partially ordered set. Every totally ordered subset of $\mathcal{S}$ has a least element: the interesection of the elements of the chain, which is non-empty by Cantor's theorem. Thus, by Zorn's lemma $\mathcal{S}$ has a minimal element M.

Now consider $T|_M : M \to M$, and let $N \subseteq M$ be a closed, non-empty $T|_M$-invariant subset. Then, N is also T-invariant, which implies $N = M$ since M is minimal. By (ii) of Proposition 1.21 we have that $T|_M$ is minimal. □

Since a minimal transformation cannot have non-trivial closed invariant sets, it will not have any finite invariants sets unless X is finite. In particular, if T is minimal and X is infinite, the dynamical system $(X, T)_{\mathbb{Z}}$ cannot have periodic points. Nonetheless, there are homeomorphisms of X with a dense set of periodic orbits and with $\{x \in X : \overline{\mathcal{O}_T(x)} = X\}$ also being dense.

The following concept is a weakening of minimality:

Definition 1.24. T is called *topologically transitive* if for some $x \in X$ we have $\overline{\mathcal{O}_T(x)} = X$.

Definition 1.25. Let $x \in X$. The ω-limit set of x consists of all the limit points of $\mathcal{O}_T^+(x) = \{T^n x : n \geq 0\}$, that is

$$\omega(x) = \{y \in X : \exists n_i \to \infty \text{ with } T^{n_i} x \to y\}.$$

1.2.2 Measurable dynamics

Definition 1.26. Let $(X, \mathcal{M}, \mu)$ be a probability space. A measure-preserving transformation T of $(X, \mathcal{M}, \mu)$ is called *ergodic* if the only members $E \in \mathcal{M}$ with $T^{-1}E = E$ satisfy $m(E) = 0$ or $m(E) = 1$.

Intuitively, ergodicity captures the notion of irreducibility of measurable transformations. Suppose $T : X \to X$ is a measure-preserving map and $T^{-1}E = E$, for $E \in \mathcal{M}$. Then, $T^{-1}(X \setminus E) = X \setminus E$ and we could study T by the two simpler transformations $T|_E$ and $T|_{X\setminus E}$. If $0 < m(E) < 1$, this has simplified the analysis of T. However, if $m(E) = 0$, we can ignore E and we have not significantly simplified T since neglecting a set of zero measure is allowed in measure theory.

An important piece of terminology is that if $T : X \to X$ is a continuous transformation of a compact metric space, we say $\mu \in M(X, T)$ is ergodic if the measure-preserving transformation T of the measure space $(X, \mathcal{B}(X), \mu)$ has the corresponding property.

Note that $M(X, T)$ is a compact convex subset of $M(X)$ (see [Wal00, Theorem 6.10]). This allows us to speak about its extreme points, which are the measures $\mu \in M(X)$ for which there do not exist two distinct measures $\nu, \eta \in M(X)$ and a scalar $\lambda \in (0, 1)$ such that $\mu = \lambda\nu + (1 - \lambda)\eta$.

Proposition 1.27 ([RF10, Theorem 13, §22.4])**.** *Let $T : X \to X$ be a continuous mapping on a compact metric space X. If a measure $\mu \in M(X, T)$ is an extreme point of $M(X, T)$ then μ is ergodic.*

Proof. Assume $\mu \in M(X, T)$ is not ergodic. Then, there is a Borel subset $A \subset X$ such that $T^{-1}A = A$ and $0 < \mu(A) < 1$. Define

$$\nu(E) = \frac{\mu(E \cap A)}{\mu(A)}, \quad \forall E \in \mathcal{B}(X),$$

and

$$\eta(E) = \frac{\mu(E \cap (X \setminus A))}{\mu(X \setminus A)}, \quad \forall E \in \mathcal{B}(X).$$

Then, since $\mu(X) = 1$, we have $\mu = \lambda\nu + (1 - \lambda)\eta$, where $\lambda = \mu(A)$. Both $\nu, \eta \in M(X)$ and as $\mu \in M(X, T)$, for each $E \in \mathcal{B}(X)$

$$\mu(E \cap A) = \mu(T^{-1}(E \cap A)) = \mu(T^{-1}E \cap T^{-1}A) = \mu(T^{-1}E \cap A).$$

Therefore, $\nu \in M(X, T)$ and by a similar argument $\eta \in M(X, T)$, which implies μ is not an extreme point of $M(X, T)$. Therefore, μ is ergodic. □

The converse of Proposition 1.27 holds as well: if a measure $\mu \in M(X, T)$ is ergodic then it is an extreme point of $M(X, T)$, but we will not need it.

Theorem 1.28 ([RF10, Theorem 14, §22.4])**.** *Let $T : X \to X$ be a continuous mapping on a compact metric space X. Then there is a Borel measure $\mu \in \mathcal{B}(X)$ with respect to which T is ergodic.*

We do not include proof of Theorem 1.28 because it makes use of several functional analysis results not relevant here. The next theorem relates the idea of measure theoretic dense orbits to the topological notion of dense orbits for continuous maps.

Theorem 1.29 ([Wal00, Theorem 5.15])**.** *Let $T : X \to X$ be a homeomorphism of the compact metric space X and let $\mu \in M(X,T)$ be ergodic, such that $\mu(U) > 0$ for every non-empty open set U. Then, $\mu\left(\{x \in X : \overline{\mathcal{O}_T(x)} = X\}\right) = 1$. In particular, T is topologically transitive.*

Proof. Let $(U_n)_{n\geq 1}$ be a countable basis for the topology. Then

$$\{x \in X : \overline{\mathcal{O}_T(x)} = X\} = \bigcap_{n=1}^{\infty} \bigcup_{k=-\infty}^{\infty} T^k U_n.$$

For each n the open set $\bigcup_{k=-\infty}^{\infty} T^k U_n$ is T-invariant so by ergodicity has measure 0 or 1. But U_n is contained in this set and $\mu(U) > 0$ since U_n is open. Therefore $\mu\left(\bigcup_{k=-\infty}^{\infty} T^k U_n\right) = 1$, which implies $\mu\left(\{x \in X : \overline{\mathcal{O}_T(x)} = X\}\right) = 1$. □

Definition 1.30. A continuous transformation $T : X \to X$ of a compact metrisable space X is called *uniquely ergodic* if there is only one T-invariant Borel probability measure on X, i.e., $M(X,T)$ consists of one point.

If T is uniquely ergodic and $M(X,T) = \{\mu\}$ then, by Proposition 1.27, μ is ergodic because it is an extreme point of $M(X,T)$. Moreover, unique ergodicity is connected, but no equivalent, to minimality (see [Bos85]).

Theorem 1.31 ([Wal00, Theorem 6.17])**.** *Suppose $T : X \to X$ is a homeomorphism of the compact metrisable space X. Suppose T is uniquely ergodic and $M(X,T) = \{\mu\}$. Then T is minimal if and only if $\mu(U) > 0$ for all non-empty open sets U.*

Proof. ($\Rightarrow$) Suppose T is minimal. If U is open and non-empty, by Proposition 1.21(iii) $X = \bigcup_{n=-\infty}^{\infty} T^n U$, so if $\mu(U) = 0$ then $\mu(X) = 0$, a contradiction.

($\Leftarrow$) Suppose $\mu(U) > 0$ for all open non-empty U. Suppose T is not minimal. There exists a closed set K such that $TK = K$ and $\varnothing \neq K \neq X$. The homeomorphism $T|_K$ has an invariant Borel probability measure μ_K on K by Corollary 1.15. Define $\tilde{\mu}$ on X by $\tilde{\mu}(B) = \mu_K(K \cap B)$ for all $B \in \mathcal{B}(X)$. Then $\tilde{\mu} \in M(X,T)$ and $\tilde{\mu} \neq \mu$ because $\mu(X \setminus K) > 0$, as $X \setminus K$ is non-empty and open, while $\tilde{\mu}(X \setminus K) = 0$. This contradicts the unique ergodicity of T. □

Chapter 2

Billiard tables

In this chapter, we introduce general billiard tables within an appropriate class of domains and begin our discussion about rational billiards. We answer Q3 within the framework of topological dynamics: we prove in §2.2.2 that billiard trajectories are typically dense in the configuration space (Theorem 2.18) and orbits are not even topologically transitive in the phase space (Theorem 2.12).

The primary reference for the general exposition and notation about billiards is [Che14]. The main results of this chapter are from [BKM78], as well as the crucial relation between rational billiards and interval exchange transformations.

2.1 Basic constructions

Consider a bounded domain $\mathcal{P}_0 \subset \mathbb{C}$ and let $\mathcal{P} = \overline{\mathcal{P}_0}$ denote its closure. Let us describe some convenient assumptions on the topological boundary $\partial\mathcal{P}$ to avoid pathological situations, which are tacitly excluded in most research articles of the literature (cf. [Che14]).

Assumption A1. The boundary $\partial\mathcal{P}$ is a finite union of C^ℓ-smooth compact curves, with $\ell \geq 3$. Hence, $\Gamma := \partial\mathcal{P} = \Gamma_1 \cup \cdots \cup \Gamma_r$, with $r \in \mathbb{N}$, where each Γ_i is defined by a C^ℓ-map $\gamma_i : [a_i, b_i] \to \mathbb{C}$ parametrized by arclength, which is one-to-one on $[a_i, b_i)$ and has one-sided derivatives up to order ℓ at the points a_i and b_i.

Assumption A2. The boundary components Γ_i only intersect at their endpoints, that is $\Gamma_i \cap \Gamma_j \subset \partial\Gamma_i \cup \partial\Gamma_j$, for $i \neq j$.

We will call $\Gamma_* := \partial\Gamma_1 \cup \cdots \cup \partial\Gamma_r$ the *vertices* of $\mathcal{P}$ and $\tilde{\Gamma} := \Gamma \setminus \Gamma_*$ the *regular boundary points* of $\mathcal{P}$. Chernov and Markarian [Che14, §2] impose stricter restrictions on Γ in order to deal with delicate situations which occur in more general domains. Nevertheless, these will not be needed for our purposes since we will not treat curved boundaries.

Definition 2.1. A *billiard table* is the closure $\mathcal{P}$ of a bounded domain $\mathcal{P}_0 \subset \mathbb{C}$ whose boundary satisfies A1 and A2.

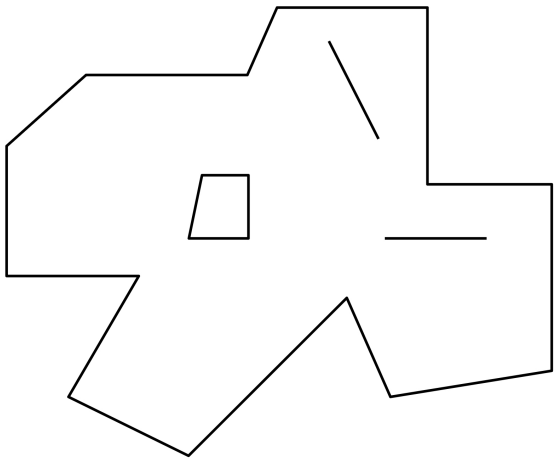


Figure 2.1: Billiard table with a hole and two slits.

Note that from this definition we are ruling out *thin* holes within $\mathcal{P}$. More precisely, if $\mathcal{P}_0$ is a bounded domain with a segment removed from its inside, the closure $\mathcal{P}$ will absorb it. Nevertheless we may add finitely many slits as desired as in Figure 2.1 .

2.1.1 Billiard flow and phase space

Definition 2.2 (cf. [Che14])**.** A *billiard system* corresponds to a free motion of a point particle inside a billiard table $\mathcal{P}$ with specular reflections off the regular boundary $\tilde{\Gamma}$.

Let $q \in \mathcal{P}$ denote the position of the moving particle and $v \in S^1$ its velocity vector. Of course, $q = q(t)$ and $v = v(t)$ are functions of time $t \in \mathbb{R}$. When the particle moves inside the table, so that $q \in \mathcal{P}_0$, it maintains a constant velocity,

$$\dot{q} = v \quad \text{and} \quad \dot{v} = 0.$$

When the particle collides with the regular part of the boundary, $q \in \tilde{\Gamma}$, its velocity vector instantaneously gets reflected across the tangent line to $\tilde{\Gamma}$ at q (see Figure 2.2),

$$v^+ = v^- - 2\langle v, n\rangle n, \tag{2.1}$$

where v^+ and v^- refer to the postcollsional and precollisional velocities, respectively, and n denotes the unit normal vector to $\tilde{\Gamma}$ at q. Otherwise, if the particle hits a vertex, $q \in \Gamma_*$, it stops and its motion will no longer be defined beyond that point. Therefore, the trajectory of the particle is defined at all times $t \in \mathbb{R}$ unless

(a) The particle hits a vertex of $\mathcal{P}$, i.e. $q(t) \in \Gamma_*$ for some $t \in \mathbb{R}$.

(b) Collision times $\{t_n\}$ have an accumulation point in $\mathbb{R}$ (see [Che14, §2.4]).

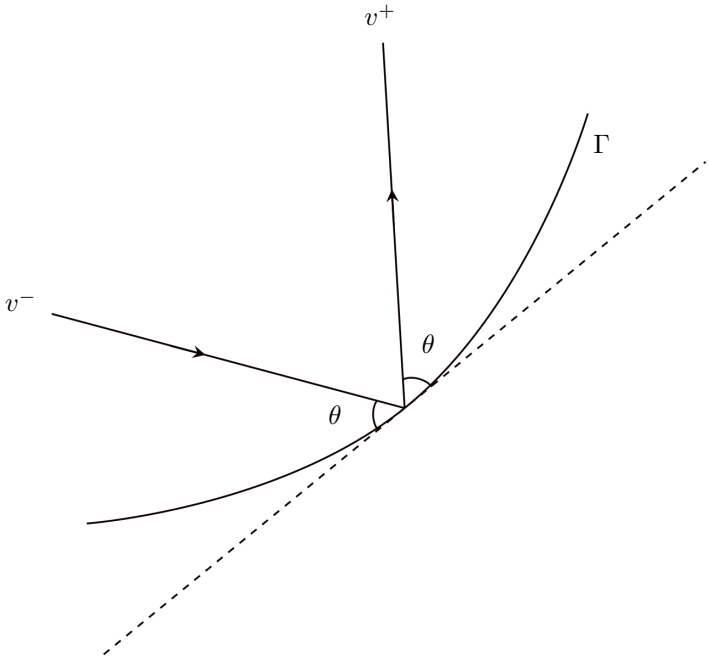


Figure 2.2: Behavior of a billiard trajectory before (v^-) and after (v^+) a collision with Γ.

The state of the moving particle at any time is completely specified by its position $q \in \mathcal{P}$ and velocity $v \in S^1$. As a consequence, the natural phase space for the system is $\Omega = \mathcal{P} \times S^1/\sim$, where at each regular boundary point $q \in \tilde{\Gamma}$ we identify pairs $(q, v^+) \sim (q, v^-)$ related by eq. (2.1). This amounts to gluing Ω along its boundary $\partial\Omega = \Gamma \times S^1$.

Let $\tilde{\Omega} \subset \Omega$ be the set of states (q, v) on which the dynamics of the moving particle is defined at all times $t \in \mathbb{R}$. We obtain a one-parameter group of transformations $\Phi^t : \tilde{\Omega} \to \tilde{\Omega}$, which induces the *billiard flow*

$$\Phi : \tilde{\Omega} \times \mathbb{R} \longrightarrow \tilde{\Omega}. \tag{2.2}$$

As a result, we have a dynamical system $(\tilde{\Omega}, \Phi)$ on $(\mathbb{R}, +)$, from which we are interested in the asymptotic measurable and topological properties of its orbits.

Denote π_1 and π_2 the natural projections of Ω onto $\mathcal{P}$ and S^1, respectively. Every orbit of the flow is a continuous curve in Ω, endowed with the quotient topology coming from the product (the continuity at collisions is ensured by our identification of v^+ with v^-).

Definition 2.3. Given $x \in \tilde{\Omega}$, we call $\mathrm{Tr}(x) := \{\pi_1(\Phi^t x) : t \in I\} \subset \mathcal{P}$ a *billiard trajectory*, where I is an interval of the real line.

Distinction between orbits and trajectories raises the important difference between dynamics on the phase space versus dynamics on the configuration space. In the Introduction we posed several questions, which now may be subtly refined with this observation. For

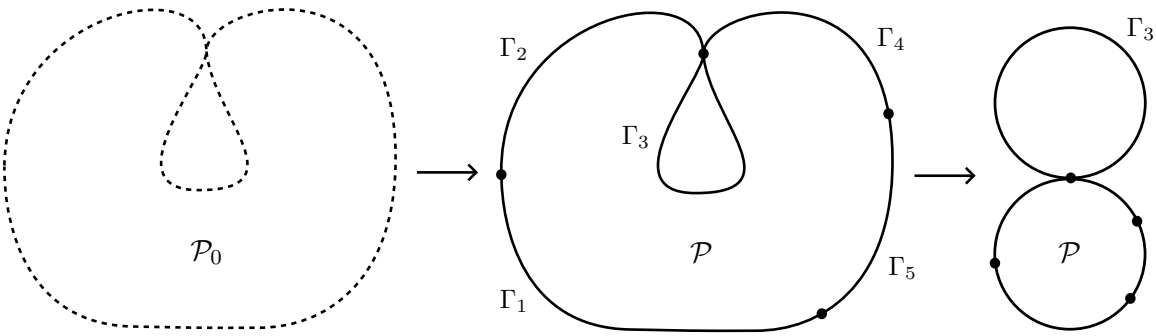


Figure 2.3: Homotopy equivalence with the wedge of two circles after passing to the closure.

instance, question Q3 presents opposed answers in relation to the density of the orbits in Ω, in contrast with the density of trajectories $\mathrm{Tr}(x)$ in $\mathcal{P}$, as will be shown in §2.2.2.

Finally, we derive a map from the flow (2.2), as in §1.2 (see eq. (1.4)). Let $B = \bigcup_i B_i$, where $B_i := \{x = (q, v) : q \in \Gamma_i,\ \langle v, n_i\rangle \geq 0\} \subset \partial\Omega$ and n_i is the unit normal vector to Γ_i at q, pointing inside $\mathcal{P}$. We define the *billiard map* ([BKM78]; cf. [Che14])

$$\mathcal{F} : B \longrightarrow B \quad \text{by} \quad \mathcal{F}(x) = \Phi^{\tau(x)}x, \tag{2.3}$$

where $\tau(x) = \inf\{t > 0 : \Phi^t x \in \partial\Omega\}$ is the first return time to $\tilde{\Gamma}$. Let $\tilde{B} = B \cap \tilde{\Omega}$, i.e. the boundary points with trajectories defined for all times and initial direction pointing inwards. Then, $\mathcal{F}|_{\tilde{B}}$ is continous and invertible.

2.1.2 Topology of the billiard class

Although the problem of dynamics in billiards is essentially of geometric nature, topological subtleties may cause major complications if they are not treated carefully. Condition A1 excludes badly behaved components such as the topologist's sine curve $\gamma(t) = t + i\sin\left(\frac{1}{t}\right)$ near $t = 0$, while condition A2 forces Γ to be cut into appropriate pieces. We are going to see that the boundary of a billiard table is connected. Actually, it has the same homotopy type as a wedge of circles (see Figure 2.3), but we will not dwell on that.

First, consider $p \in \Gamma$, then p is either a regular boundary point or a vertex. In the first case, by A2 we have that $p \in \Gamma_i$ for a unique $i \in \{1, \ldots, r\}$, and by continuity of γ_i it is clear that we have a connected open neighborhood of p. In the second case, suppose $p \in \partial\Gamma_{i_1} \cap \cdots \cap \partial\Gamma_{i_{m_p}}$, with $1 \leq i_k \leq r$ and m_p being the number of sides that contain the vertex p. Again, by A2 we can choose a small enough open neighborhood $U \subset \mathbb{C}$ of p such that $U \cap \Gamma$ is the union of m_p disjoint arcs having p as its common point. Since the union of a collection of connected subspaces that have a point in common is connected [Mun14, Theorem 23.3], we conclude that Γ is locally connected.

Secondly, suppose that $\mathcal{P}_0 \subset \mathbb{C}$ is a simply connected bounded domain and assume, without loss of generality, that $0 \in \mathcal{P}_0$. By the Riemann mapping theorem (see [Ahl07])

there exists a unique conformal isomorphism

$$\phi : D \longrightarrow \mathcal{P}_0 \tag{2.4}$$

such that $\phi(0) = 0$, where $D = \{z \in \mathbb{C} : |z| < 1\}$. Luo and Yao provide in [LY22] a thorough analysis of the extension theorems for the map (2.4). In particular, the Carathéodory-Torhorst theorem states that ϕ has a continuous extension

$$\hat{\phi} : \overline{D} \longrightarrow \mathcal{P} \tag{2.5}$$

if and only if Γ is locally connected. This implies the restriction $\Phi = \hat{\phi}|_{S^1} : S^1 \to \Gamma$ must be continuous and therefore Γ is connected.

Accordingly, $\Gamma \subset \mathbb{C}$ is a connected, locally connected subset of the Euclidean plane separating 0 from ∞ in the Riemann sphere $S^2 = \mathbb{C} \cup \{\infty\}$. Then $S^2 \setminus \Gamma$ consist of at least two components; all complementary components are simply connected.

2.2 Rational billiards

Definition 2.4. A *polygonal billiard* is a billiard table such that each boundary component Γ_i has zero curvature. The *walls* or *sides* of a polygonal billiard are the (flat) boundary components of Γ.

Let $\mathcal{P}$ be a polygonal billiard with n sides. Let $G = \langle \sigma_1, \ldots, \sigma_n \rangle \leq O(2)$ be the group generated by the linear part of the reflections across the sides of $\mathcal{P}$, that is, the reflections across the lines through the origin which are parallel to the sides of the polygonal billiard $\mathcal{P}$ (see Figure 2.4).

Definition 2.5 ([Vor96])**.** A polygonal billiard $\mathcal{P}$ is called *rational* if the group G is finite and *irrational* otherwise.

Every reflection satisfies $\sigma_i^2 = 1$, so G is a finitely generated group with generators of order 2. It is somehow counterintuitive that G may not be finite. This is the well-known Burnside Problem in the theory of groups (see [Hal18, §18]).

Proposition 2.6. *Let $\mathcal{P}$ be a simply connected polygonal billiard. Then $\mathcal{P}$ is rational if and only if its angles are rational multiples of π.*

Proof. Since $\mathcal{P}$ is simply connected, §2.1.2 ensures the boundary is connected, so the angles between any two lines that generate the (linear parts of the) reflections of G are necessarily rational.

($\Rightarrow$) Assume $\mathcal{P}$ is rational. Let $\sigma_i, \sigma_{i+1} \in G$ be reflections corresponding to two adjacent sides of $\mathcal{P}$ meeting at an angle θ. It is well-known that $r = \sigma_1\sigma_2 \in G$ is a rotation of angle 2θ. Since G is finite, there must exist $n \in \mathbb{N}$ such that $r^n = \mathrm{Id}$. Notice that r^n is a rotation of angle $2\theta n$, but the only rotation that is the identity is a rotation by an integer multiple of 2π. This forces $2\theta n = 2\pi k$, $k \in \mathbb{Z}$ and therefore, $\theta = \frac{k}{n}\pi$.

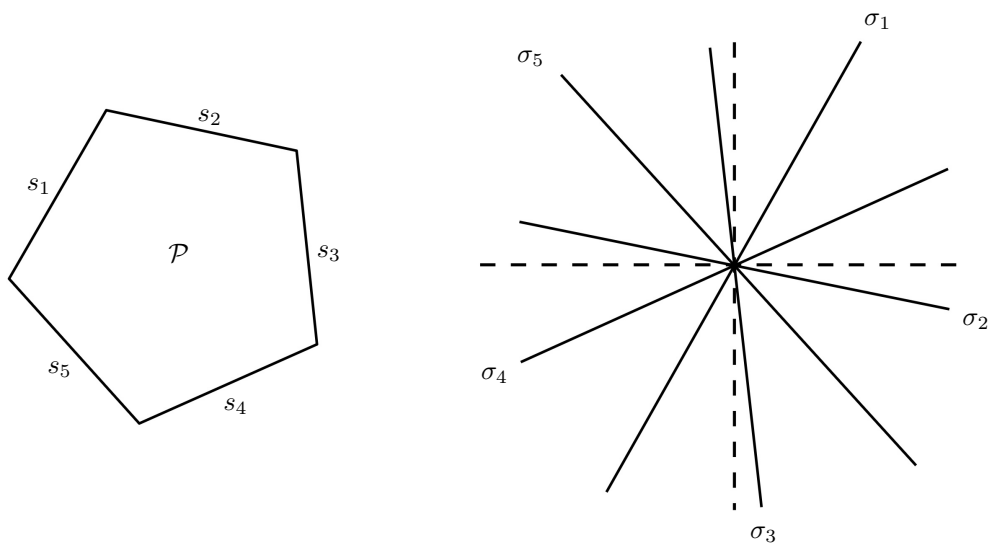


Figure 2.4: Orientation-reversing isometries $\sigma_1, \ldots, \sigma_5$ generated by the linear parts of the reflections across the sides $s_1, \ldots, s_5$ of the billiard $\mathcal{P}$.

($\Leftarrow$) Assume all angles are rational multiples of π. Let $R = G \cap SO(2)$ be the subgroup of rotations of G. Let $\sigma_i, \sigma_j \in G$ be reflections corresponding to sides of $\mathcal{P}$ making an angle $\theta_{ij} = \frac{p_{ij}}{q_{ij}}\pi$, with $p_{ij}, q_{ij} \in \mathbb{Z}$ coprime and $q_{ij} \neq 0$. Again, we have that $r_{ij} = \sigma_i\sigma_j$ is a rotation of angle $2\theta_{ij}$ and order

$$m_{ij} = \frac{q_{ij}}{\gcd(2p_{ij}, q_{ij})}.$$

Let N be the least common multiple of the orders m_{ij}, $1 \leq i, j \leq n$. We make the canonical identification $SO(2) \cong \left\{e^{i\theta} : \theta \in \mathbb{R}\right\} \subset \mathbb{C}$ and consider the subgroup of N-th roots of unity $\mu_N = \left\{e^{2\pi ik/N} : k = 0, 1, \ldots, N-1\right\}$, which is finite by the fundamental theorem of algebra. Since

$$R = \langle r_{ij} : 1 \leq i \leq j \leq n \rangle \quad \text{and} \quad r_{ij}^N = \left(r_{ij}^{m_{ij}}\right)^{N/m_{ij}} = 1,$$

we have that each $r_{ij} \in \mu_N$, therefore $R \leq \mu_N$ and R is finite.

Moreover, notice that $R \trianglelefteq G$ since conjugation preserves the determinant. Consider the homomorphism $\det : O(2) \to \{\pm 1\}$ and restrict it to G as follows

$$\det|_G : G \longrightarrow \{\pm 1\}.$$

Clearly, $\det|_G$ is surjective and $\ker(\det|_G) = R$. Thus, by the First Isomorphism Theorem, $G/R \cong \{\pm 1\}$, which implies that $[G : R] = 2$. By Lagrange's Theorem $|G| = [G : R]\,|R| = 2|R|$. Thus, G is finite. $\square$

Remark 2.1. Many texts define rational billiards as planar polygons whose angles are rational multiples of π (e.g. [ZK75],[BKM78] or [Gut84]) as opposed to [Vor96]. However, if simple-connectedness is not assumed, Proposition 2.6 is no longer true. For instance, consider the billiard $\mathcal{P} = [-1, 1]^2$ and remove a small square from inside, rotated by an angle of $\sqrt{2}\pi$ with respect to the x-axis. Then, although all the angles between adjacent sides are $\frac{\pi}{2}$, the group G contains an irrational rotation and therefore is infinite.

2.2.1 Interval exchange transformations

We need to introduce the skeleton of simple yet very rich dynamical systems called interval exchange transformations (IETs). These maps were first described in [Kea75] and have been intensively studied (see [Vee78], [Mas82], [Bos85]; cf. [Via06],[Yoc10]). We will see in §2.2.2 how they are intimately connected to rational billiards.

Let $n \geq 2$ and $\Lambda_n = \{\lambda = (\lambda_1, \ldots, \lambda_n) \in \mathbb{R}^n \ : \ \lambda_i > 0\}$. For $\lambda \in \Lambda_n$, set the *separation points*

$$\beta_i = \begin{cases} 0 & \text{if } \ i = 0 \\ \sum_{j=1}^{i} \lambda_i & \text{if } \ 1 \leq i \leq n. \end{cases} \tag{2.6}$$

For $1 \leq i \leq n$, set the intervals $X_i = [\beta_{i-1}, \beta_i)$ and $X = \bigcup_{i=1}^{n} X_i = [0, \beta_n)$. Let Σ_n be the permutation group of $\{1, 2, \ldots, n\}$.

Given $\lambda \in \Lambda_n$ and $\tau \in \Sigma_n$, let $\lambda^\tau = (\lambda_{\tau^{-1}(1)}, \lambda_{\tau^{-1}(2)}, \ldots, \lambda_{\tau^{-1}(n)})$. Analogously to (2.6), form the corresponding

$$\beta_i^\tau = \begin{cases} 0 & \text{if } \ i = 0 \\ \sum_{k=1}^{i} \lambda_{\tau^{-1}(k)} & \text{if } \ 1 \leq i \leq n \end{cases}$$

and $X_i^\tau = [\beta_{i-1}^\tau, \beta_i^\tau)$, and consider $T : X \to X$ defined by

$$T(x) = x - \beta_{i-1} + \beta_{\tau(i)-1}^\tau, \quad x \in X_i, \quad 1 \leq i \leq n. \tag{2.7}$$

The map (2.7) is called the (λ, τ)-*interval exchange transformation* and it is sometimes denoted simply as $T_{\lambda,\tau}$ or T, provided the choice of (λ, τ) is clear. It exchanges the subintervals X_i according to the permutation τ, namely $T(X_i) = X_{\tau(i)}^\tau$ (see Figure 2.5). Moreover, T is continuous, except perhaps at the points β_i, $1 \leq i \leq n-1$, where it is right-continuous. Observe that interval exchange transformations are the piecewise order-preserving isometries of X, with an inverse map given by $(\lambda^\tau, \tau^{-1})$. All in all, from the dynamical perspective, we want to study the asymptotic behavior of the orbits of $(X, T_{\lambda,\tau})_{\mathbb{Z}}$.

Definition 2.7. We call a permutation $\tau \in \Sigma_n$ *irreducible* if for each $1 \leq j \leq n-1$,

$$\tau(\{1, 2, \ldots, j\}) \neq \{1, 2, \ldots, j\}.$$

If τ is not irreducible, then obviously T can be decomposed into an interval exchange transformation on $[0, \beta_j)$ and one on $[\beta_j, \beta_n)$. Thus, we can assume subsequently, with no loss of generality, that τ is irreducible.

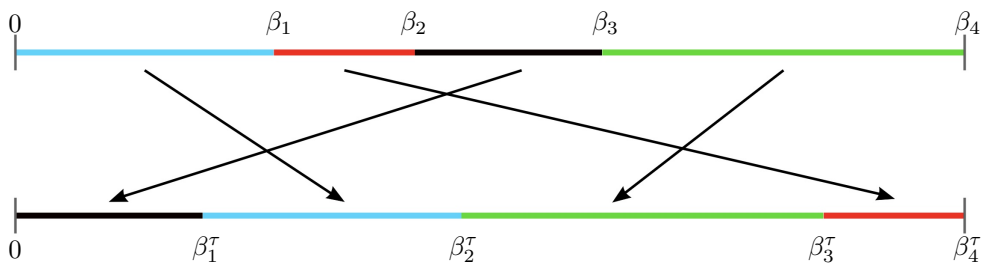


Figure 2.5: Interval exchange transformation with permutation $\tau : (1, 2, 3, 4) \mapsto (3, 1, 4, 2)$.

Definition 2.8 ([Kea75]; cf. [Vee78])**.** An interval exchange transformation $T_{\lambda,\tau}$ is called *interval-prime* if

M1. T is aperiodic, or equivalently, $\mathcal{O}_T(x)$ is infinite for every $x \in X$.

M2. If F is a finite union of half-open intervals whose endpoints all belong to the countable set

$$D^\infty = \bigcup_{i=0}^{n-1} \mathcal{O}_T(\beta_i) \cup \{\beta_n\},$$

then $TF = F$ implies $F = X$ or $F = \varnothing$.

We must adapt the concept of minimality from §1.2.1, which was originally defined for homeomorphisms of compact metric spaces. Within the IETs framework, we say that T is *minimal* if $\mathcal{O}_T(x)$ is dense in X, for every $x \in X$. In [Kea75, §5], Keane provides a more fitting space in which interval exchange transformations become homeomorphisms, but it will not be useful for our purposes.

Theorem 2.9 ([Kea75, p. 26])**.** *T is interval-prime if and only if T is minimal.*

Proof. ($\Leftarrow$) Suppose T is not interval-prime. Then, there exist either periodic points or there exists F as in M2 such that $TF = F$ and $\varnothing \neq F \neq X$, in which case $\mathcal{O}_T(x) \subset F$ for any $x \in F$. Therefore, $\overline{\mathcal{O}_T(x)} \subset \overline{F} \neq X$ by the construction of F.

($\Rightarrow$) Suppose T is interval-prime and let $x \in X$ such that the open set $U = X \setminus \overline{\mathcal{O}_T(x)}$ is non-empty. Then, there exists a half-open interval I of positive length, contained in U such that $I \cap \mathcal{O}_T(x) = \varnothing$, or equivalently, $x \notin T^r I$ for each $r \in \mathbb{Z}$. It suffices now to show that for any half-open interval I of positive length, there exists an integer k such that $\bigcup_{r=0}^{k} T^r I = X$.

Let $I = [a, b)$. Since D^∞ is countable, we can assume that $a \notin D^\infty$ and $b \notin D^\infty$, by reducing I if necessary.

For each $y \in D^\infty \cup \{a, b\}$, set

$$k(y) = \inf \left\{ t \geq 0 : T^{-t} y \in (a, b) \right\}.$$

The points $\left\{ T^{-k(y)} y : y \in D^\infty \cup \{a, b\},\ k(y) < \infty \right\}$ partition I into half-open intervals $I_1, I_2, \ldots, I_\ell$, with $\ell \leq n + 2$. Fix j with $1 \leq j \leq \ell$ and let m be minimal such that $T^m I_j \cap I \neq \varnothing$. Then by construction of the partition of I, each of the sets $I_j, TI_j, \ldots, T^m I_j$ is a single half-open interval: if not, then for some $0 \leq s \leq m - 1$, $T^s \mathring{I}_j$ would contain a point $y \in D^\infty$. If s is minimal with this property, then it follows that $k(y) = s$ and $T^{-k(s)} y \in \mathring{I}_j$, contrary to the choice of the partition.

The same reasoning applied to the endpoints a and b of I implies that $T^m \mathring{I}_j$ does not contain a or b, i.e. $T^m I_j \subseteq I$. Thus, for each j there exists $m_j = m$ such that the intervals $I_j, TI_j, \ldots, T^{m_j - 1} I_j$ are disjoint with each other and with I, and $T^{m_j} I_j \subseteq I$. It follows that

$$F = \bigcup_{j=1}^{\ell} \bigcup_{i=0}^{m_j - 1} T^i I_j$$

is a finite union of intervals and $TF = F$. Moreover, if x is a boundary point of F not belonging to D^∞, then $T^r x$ is a boundary point of F for each t, since each T^r is continuous on $X \setminus D^\infty$. This yields a contradiction, since according to M1, $\mathcal{O}_T(x)$ is infinite, but F as a finite union of intervals has only finitely many boundary points. Then M2 implies that $F = X$. □

Corollary 2.10. *For each interval of positive length, the return time (see §1.2) to this interval is bounded.*

Proposition 2.11 ([Kea75])**.** *If the orbits of the separation points $D = \{\beta_1, \ldots, \beta_{n-1}\}$ are infinite and distinct, then T is interval-prime.*

Proof. Suppose that $x \in X$ and $t \geq 0$ with $T^t x = x$. We set

$$\beta = \max\{T^i \beta_j : 0 \leq i \leq t - 1,\ 0 \leq j \leq n - 1,\ T^i \beta_j \leq x\}.$$

If $\beta = x$, then either one of the points of D is periodic, or $x = T^i 0$ for some i. Now 0 is in TD unless $\tau(1) = 1$, and $\tau(1)$ implies that $\beta_1 \in TD$, which contradicts our hypothesis. Thus, $x = T^{i+1} \beta_j$ for some $1 \leq j \leq n - 1$, and this β_j is periodic. So we must have $\beta < x$. But each T^i with $1 \leq i \leq t$ is an isometry on $[\beta, x]$, so that $T^t \beta = \beta$, which is a contradiction, and we have M1.

Let F be as in condition M2. If $x \neq \beta_n$ is a boundary point of F, then either Tx is a boundary point of F or $x \in D \cup \{0\}$. Since F possesses a finite number of boundary points, there exists a non-negative integer t such that $T^t x \in D \cup \{0\}$. Likewise, either $T^{-1} x$ is a boundary point or $T^{-1} x \in D$, and there exists a negative integer s with $T^s x \in D$. Then, $y = T^{t-s} y \in D \cup \{0\}$, which contradicts our hypothesis unless $t = 0$, $s = -1$ and $x = 0$. Note that $0 \in TD$. Therefore $F = X$ and M2 is satisfied. □

2.2.2 Minimality of billiard trajectories

Throughout this part, we will be working with the canonical identification $S^1 \cong \mathbb{R}/2\pi\mathbb{Z}$, so for $v \in S^1$ we will be dealing with the counterclockwise angle $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ that v makes with the x-axis.

Let $\mathcal{P}$ be a rational billiard with sides $a_1, \ldots, a_n$. Consider the action induced by G on S^1 as follows

$$\begin{aligned} G \times S^1 &\longrightarrow S^1 \\ (g, \theta) &\longmapsto g(\theta), \end{aligned} \tag{2.8}$$

where $g = \sigma_{i_1} \cdots \sigma_{i_k}$, $1 \leq i_1, \ldots, i_k \leq n$. In Proposition 2.6 we proved that G must be finite, so the orbits of the action (3.2), denoted by $\mathrm{Orb}_G(\theta) = \{g(\theta) : g \in G\}$, must be finite.

Theorem 2.12. *The billiard flow $(\tilde{\Omega}, \Phi)$ is not topologically transitive.*

Proof. Given initial data $x = (q, \theta) \in \tilde{\Omega}$, consider the subset $Q = (\mathcal{P} \times \mathrm{Orb}_G(\theta))/\sim$ of Ω. Since $\mathrm{Orb}_G(\theta)$ is finite, then Q is closed in Ω. Let $\tilde{Q} = Q \cap \tilde{\Omega}$, which is closed in $\tilde{\Omega}$. Clearly, $x \in \tilde{Q}$, so $\tilde{Q} \neq \varnothing$. Also, $S^1 \setminus \mathrm{Orb}_G(\theta)$ is non-empty, and therefore, $\tilde{Q}$ is a proper subset of $\tilde{\Omega}$. The key fact is that $\tilde{Q}$ is an invariant set for the billiard flow $\Phi^t : \tilde{\Omega} \to \tilde{\Omega}$. Suppose there exists $t \in \mathbb{R}$ such that $\Phi^t x \notin \tilde{Q}$. Then, since $\Phi^t x \in Q$ we have necessarily that $x \notin \tilde{\Omega}$, contradicting our choice. Now, $\tilde{\Omega} \setminus \tilde{Q}$ is a non-empty open set which does not intersect the Φ-orbit of x. Hence, $(\tilde{\Omega}, \Phi)$ cannot be topologically transitive. □

Remark 2.2. Notice that the initial data $x = (q, \theta)$ is really an equivalence class $[(q, \theta)]$ which consists of only one element (q, θ) if $q \in \mathcal{P} \setminus \Gamma$, but it has two if $q \in \Gamma$, by definition of Ω.

Nevertheless, the matter is completely different in the case of billiard trajectories, as we briefly mentioned in §2.1.1. Let $\Gamma_* = \{A_1, \ldots, A_n\}$ be the set of vertices of $\mathcal{P}$, such that

$$\begin{aligned} A_i &= \angle(a_i, a_{i+1}), \quad 1 \leq i \leq n-1, \\ A_n &= \angle(a_n, a_1). \end{aligned}$$

Definition 2.13. Let $A_i, A_j \in \Gamma_*$ with $j \notin \{i-1, i, i+1\}$. Fix $L \in \mathbb{N}$. A *generalized diagonal of length L* is a billiard trajectory connecting A_i with A_j that satisfies:

(i) The number of (intermediate) collisions between A_i and A_j is exactly L.

(ii) No intermediate vertex is hit along the trajectory.

Proposition 2.14. *For a fixed $L \in \mathbb{N}$, the number of generalized diagonals of length L is finite.*

Proof. For any billiard trajectory we can compose a word that keeps track of the ordered collisions of the trajectory with the sides of $\mathcal{P}$. Suppose we begin at (q, θ), we collide with a_{i_1} and after the reflection we follow a straight line trajectory until we collide with a_{i_2}.

Repeating this process, we record the collisions as a word $a_{i_1}a_{i_2}\cdots a_{i_j}\cdots$, with $1 \leq i_k \leq n$. In this way we construct a word over a finite alphabet.

Fix two vertices A_i and A_j with $i \neq j$. Since the number of intermediate collision L is fixed, every generalized diagonal between A_i and A_j corresponds to a word $a_{i_1}\cdots a_{i_L}$. There are n^L possible such words because we allow repetition and order matters. Beware, however, that not every word is admissible in the sense that a generalized diagonal starting from A_i and ending at A_j cannot have a word whose first letter is a_i or a_{i+1}, or a word whose last letter is a_j or a_{j+1}.

Nevertheless, we claim that for each admissible word, either there exists a unique angle connecting A_i to A_j or there exists none. Choose an admissible word $w = a_{i_1}\cdots a_{i_L}$ and suppose there exists one such angle $\theta > 0$ (with respect to a fix direction, say a_{i+1}). Now, suppose there exists small enough $\theta' \neq \theta$ also connecting $A_i \to A_j$ via w. We follow the two trajectories simultaneously: the first collision with a_{i_1} necessarily occurs at different points. Let us see that the collision with a_{i_2} has the same property. In fact, the distance between the simultaneous collision points is increasing.

Without loss of generality let $\theta < \theta'$ and denote ϕ and ϕ' the angles of incidence at a_{i_1} coming from A_i with angles θ and θ', respectively. We know that two half-lines diverge if the sum of the interior angles they make with a chosen direction is greater than π. Let γ be the angle that a_{i_1} makes with a_{i+1}, the initially chosen horizontal direction adjacent to A_i. By simple trigonometry (see Figure 2.6) we have that $\phi = \theta + \gamma$ and $\phi' = \theta' + \gamma$. Then, the sum of the interior angles of the reflected directions is

$$\phi' + \phi + (\pi - 2\phi) = \gamma + \theta' - \gamma - \theta + \pi = \theta' - \theta + \pi > \pi,$$

since $\theta' - \theta > 0$.

This argument applies at each collision and the distance between the collision points increases due to the divergence of the lines, so after L collisions the two trajectories cannot converge to the same point A_j, which is a contradiction.

Hence, every admissible word admits at most one angle making it into a generalized diagonal. The list of admissible words with their corresponding angle is bounded by n^L, thus the result is proved. □

Corollary 2.15. *The number of generalized diagonals is countable.*

Consider the billiard map $\mathcal{F} : B \to B$, as defined in (2.3). Without loss of generality, choose the side a_1 and an initial angle θ. Let B_0 consist of all pairs of sides and angles actually visited starting from a_1 with the predefined direction θ. We will see below that all sides are actually represented in B_0.

Denote $\mathcal{F}_0 = \mathcal{F}|_{B_0} : B_0 \to B_0$. Now, B_0 consists of a certain number of sides together with angles θ_i^j, $j = 1, \ldots, k_i$, belonging to the side a_i. If we draw side by side k_i copies of the side a_i for each i, and contract the j-th copy of a_i by a factor $\sin\theta_i^j$, then $\mathcal{F}_0$ becomes, if the intervals are correctly arranged, an interval exchange transformation T of these intervals, whose separation points β_i correspond to vertices of the original billiard table.

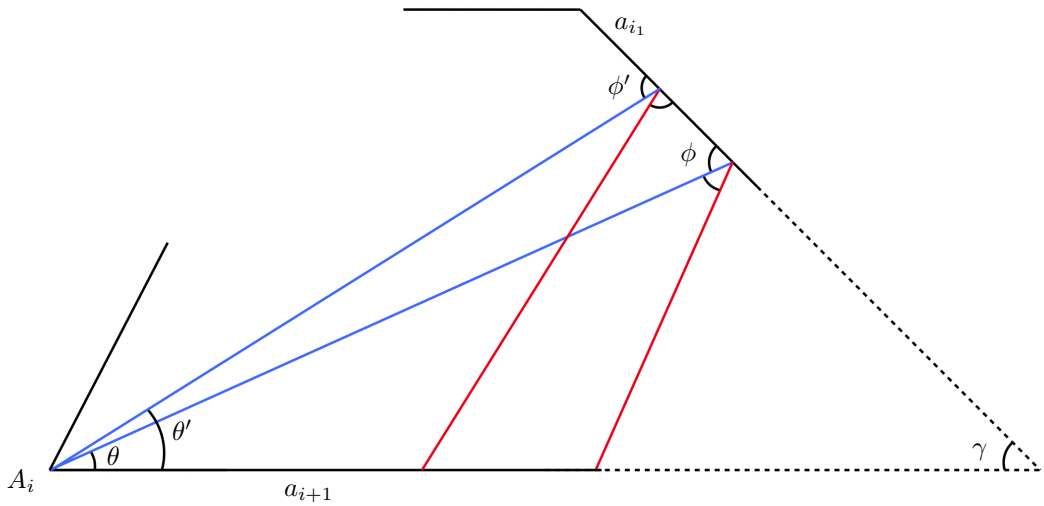


Figure 2.6: Geometric intuition behind proof of Proposition 2.14. Notice that the angle of the incident direction (blue) equal the angle of the reflected direction (red).

Definition 2.16. Let $q \in \mathcal{P}$. A direction $\theta \in S^1$ is called *exceptional* if the trajectory $\mathrm{Tr}(x)$ contains a generalized diagonal, where $x = (q, \theta)$.

When we have not fixed $q \in \mathcal{P}$ *a priori*, we may also understand an exceptional direction as $\theta \in S^1$ such that for some $q \in \mathcal{P}$, the trajectory $\mathrm{Tr}(x)$ contains a generalized diagonal, where $x = (q, \theta)$.

Lemma 2.17 (cf. [BKM78])**.** *The interval exchange transformation T generated by the billiard flow is minimal for almost all initial directions $\theta \in S^1$.*

Proof. Suppose θ is not exceptional and let us see that T is interval-prime. Since we are excluding generalized diagonals, no vertex can be periodic neither its orbit under $\mathcal{F}_0$ can be finite. Now, since the separation points of T coincide with the vertices of $\mathcal{P}$, by Proposition 2.11 we have that T is interval-prime, and thus, it is minimal. Notice that we have removed all exceptional directions; however, Corollary 2.15 ensures they are countable, and therefore, of zero measure.

□

Theorem 2.18 ([BKM78])**.** *Let $q \in \mathcal{P}$. For almost all $\theta \in S^1$, the billiard trajectory $Tr(x)$ is dense in $\mathcal{P}$, where $x = (q, \theta)$.*

Proof. From Lemma 2.17 it follows that $\overline{\mathcal{O}_{\mathcal{F}}(x)} = B_0$. However, one easily sees that if θ is not exceptional, no side can stay all the time in the shade, i.e., all sides of the polygon are

represented in B_0. Indeed, suppose that the side a_i, lying between the vertices A_{i-1} and A_i is in the shade, while some copy of a_{i-1} is in B_0. Then, points in a_{i-1} in a neighborhood of A_{i-1} are visited by the flow coming from some other side a_j with some angle θ_j^m. If all points in a_i are in the shade, then θ_j^m is a direction connecting A_{i-1} with a vertex of a_j, i.e. θ is an exceptional direction. So, if θ is not exceptional some point of a_i is visited and therefore the trajectory is dense in a_i. In a similar way one can see no internal region of the polygon can stay in the shade. Indeed, such a region would be, by definition, an open polygonal shape, whose sides would consist of segments of trajectories connecting two vertices and the conclusion follows as above. $\square$

Chapter 3

An interlude on Teichmüller theory

It is a notorious fact that minimality for interval exchange transformations does not yield unique ergodicity (see [Bos85]), with few exceptions. As a consequence, we need more powerful tools from Teichmüller theory in order to deal with the problem of unique ergodicity of the billiard flow.

In this chapter we present quadratic differentials as mathematical objects with a two-fold nature: the purely analytic perspective and the geometric approach with half-translation structures. We explore the equivalence of the two points of view, analogously to [Mas22, §2]. The main references here are [Far12], [FM15] and [Str84].

3.1 Riemann surfaces

Definition 3.1 (cf. [Far92])**.** A *Riemann surface* M is a one-complex-dimensional connected complex analytic manifold; that is, a connected, orientable topological surface S with a maximal set of charts $\{(U_\alpha, z_\alpha)\}_{\alpha\in A}$ such that

(i) The $\{U_\alpha\}_{\alpha\in A}$ constitutes an open cover of S and

$$z_\alpha : U_\alpha \longrightarrow \mathbb{C}$$

is a homeomorphism onto an open subset of the complex plane.

(ii) The *transition functions*

$$f_{\alpha\beta} = z_\alpha \circ z_\beta^{-1} : z_\beta(U_\alpha \cap U_\beta) \longrightarrow z_\alpha(U_\alpha \cap U_\beta) \tag{3.1}$$

are holomorphic whenever $U_\alpha \cap U_\beta \neq \varnothing$.

Any set of charts (not necessarily maximal) that cover S and satisfy (3.1) will be called a set of *analytic coordinate charts* or *local coordinates*.

Remark 3.1. Some authors only require the underlying topological space S to be connected, Hausdorff and orientable, and then they use Rado's theorem (see [Hub06, §1.3]) to show every Riemann surface is second countable.

Let M and N be Riemann surfaces with atlases $\{(U_\alpha, z_\alpha)\}_{\alpha\in A}$ and $\{(V_\beta, w_\beta)\}_{\beta\in B}$, respectively. Consider a map $f : M \to N$. There are two pieces of terminology that will clarify the subsequent definitions:

(i) Two charts $U_\alpha \subseteq M$ and $V_\beta \subseteq N$ are called *adapted* if $f(U_\alpha) \subseteq V_\beta$.

(ii) For any two adapted charts $U_\alpha \subseteq M$ and $V_\beta \subseteq N$, we call

$$\bar{f} := w_\beta \circ f \circ z_\alpha^{-1} : z_\alpha(U_\alpha) \longrightarrow \mathbb{C}$$

a *local representation* of f, which is well-defined.

Definition 3.2. A map $f : M \to N$ is called *holomorphic* provided for every pair of adapted charts $U_\alpha \subseteq M$ and $V_\beta \subseteq N$, the local representation $\bar{f}$ is holomorphic in the complex-analytic sense.

Note that in Definition 3.1 we have made very clear the distinction between the underlying topological manifold S and the Riemann surface M, and understand M as a *complex structure* on S. Two complex structures are considered isomorphic, and therefore indistinguishable, if there exists a *biholomorphism* between them, that is, an invertible holomorphic map with holomorphic inverse. Moreover, a complex structure naturally induces a smooth oriented structure.

We introduce the moduli space and the Teichmüller space and motivate them as deformation spaces for the complex structures on a given topological surface. There are many equivalent ways to define these spaces (see [Far12]).

Definition 3.3. The *moduli space $\mathcal{M}_g$ of Riemann surfaces of genus g* is the space of all isomorphism classes of complex structures on a closed topological surface S of genus $g \geq 1$.

Definition 3.4. Let S be a closed, oriented, smooth surface of genus $g \geq 1$. *Teichmüller space* is defined as the space of equivalence classes $\mathcal{T}_g = \{(M, f)\}/\sim$ where

- M is a complex structure on S.
- $f : S \to M$ is a *marking*, i.e., an orientation-preserving diffeomorphism that records how S is identified with M.
- $(M_1, f_1) \sim (M_2, f_2)$ if and only if there exists a biholomophism $\phi : M_1 \to M_2$ such that $\phi \circ f_1$ is isotopic to f_2.

Note that the definitions of $\mathcal{M}_g$ and $\mathcal{T}_g$ does not depend on the choice of the underlying manifold S, since the genus completely classifies closed orientable surfaces. Furthermore, it is known that in dimensions 2 and 3 the categories of smooth manifolds and topological manifolds are essentially the same, i.e., every topological surface admits a unique smooth structure up to diffeomorphism. Therefore, the genus also classifies closed smooth 2-manifolds.

Let M be a smooth manifold of genus g. Let $\mathrm{Diff}^+(M)$ denote the group of orientation-preserving diffeomorphisms and $\mathrm{Diff}_0(M)$ be the group of diffeomorphisms isotopic to the identity. Clearly, $\mathrm{Diff}_0(M) \leq \mathrm{Diff}^+(M)$. Let $f \in \mathrm{Diff}_0(M)$. There exists an isotopy $f_t \in \mathrm{Diff}^+(M)$ such that $f_0 = \mathrm{id}$ and $f_1 = f$. Let $g \in \mathrm{Diff}^+(M)$ and define gf_tg^{-1}, which is an isotopy from $gf_0g^{-1} = g \circ \mathrm{id} \circ g^{-1} = \mathrm{id}$ and $gf_1g^{-1} = gfg^{-1}$. Each $gf_tg^{-1} \in \mathrm{Diff}^+(M)$, since conjugating an orientation preserving diffeomorphism by another orientation-preserving one preserves orientation. All in all, we have shown that $\mathrm{Diff}_0(M) \trianglelefteq \mathrm{Diff}^+(M)$.

Definition 3.5. The *mapping class group* of M, denoted by Mod_g, is the group

$$\mathrm{Mod}_g = \mathrm{Diff}^+(M)/\mathrm{Diff}_0(M).$$

So far we have been referring to $\mathcal{M}_g$ and $\mathcal{T}_g$ as spaces but they have been endowed with no topology. The task of building such a topology would take us too far off the main point of this thesis; however, the big picture is the following ([FM15]; for an in-depth construction see [HM76]):

A map $f : M \to N$ is called *conformal* if f is holomorphic and has a non-vanishing derivative. Equivalently, for every pair of adapted charts $U_\alpha \subseteq M$ and $V_\beta \subseteq N$ we have that $\bar{f}'(z) \neq 0$, for every $z \in z_\alpha(U_\alpha)$. Let S be a fixed closed topological manifold of genus $g \geq 1$ and let M_0 and M_1 be non-isomorphic complex structures on S. There is no way to produce a conformal map between M_0 and M_1. However, we can try to build maps $f : M_0 \to M_1$ as nearly conformal as possible. To do so, we need a reasonable way to measure the amount of non-conformality of f. It turns out that *quasiconformal maps* give a precise notion of this quantity.

Roughly speaking, we can measure the eccentricity of infinitesimal ellipses around a point $f(x) \in M_1$ which arise as the image under f of infinitesimal circles around $x \in M_0$. If we call this eccentricity coefficient $K(f, x)$, we define

$$K(x) := \sup_{x \in M_0} K(f, x).$$

A map $f : M \to N$ is called quasiconformal if $K = K(f) < \infty$. Indeed, Teichmüller proved that there exists a quasiconformal mapping $f : M_0 \to M_1$ minimizing $K(g)$ among all quasiconformal maps $g : M_0 \to M_1$ isotopic to the identity. In the literature, any such minimizing quasiconformal map in a given isotopy class is called an *extremal map*. By using extremal quasiconformal mappings, we can naturally introduce a distance between two Riemann surface structures M_0 and M_1 by

$$d(M_0, M_1) = \frac{1}{2} \ln K(f),$$

where $f : M_0 \to M_1$ is an extremal map isotopic to the identity. The metric d is called *Teichmüller metric* on $\mathcal{T}_g$. Actually, $\mathcal{T}_g$ is not only a metric space, but it can be endowed with a complex structure such that $\dim_{\mathbb{C}} \mathcal{T}_g = 3g - 3$. The main focus of Teichmüller theory is the study of the geodesic flow associated to the Teichmüller metric on the moduli space of Riemann surfaces.

Moreover, there is an action of Mod_g on $\mathcal{T}_g$ via the markings. That is

$$\begin{aligned} \mathrm{Mod}_g \times \mathcal{T}_g &\longrightarrow \mathcal{T}_g \\ ([\phi], [(M, f)]) &\longmapsto [(M, f \circ \phi^{-1})] \end{aligned} \tag{3.2}$$

Two points in $\mathcal{T}_g$ that are related by the action of Mod_g correspond to the same Riemann surface, but with different markings. Thus, from the action (3.2) we have $\mathcal{M}_g = \mathcal{T}_g/\mathrm{Mod}_g$, which clearly induces a topology on $\mathcal{M}_g$. Nevertheless, the action $\mathrm{Mod}_g \curvearrowright \mathcal{T}_g$ is properly discontinuous but not free, since some points have nontrivial stabilizers. Therefore, in contrast with Teichmüller space, the moduli space of Riemann surfaces is not in general a manifold, but an orbifold.

All the previous discussion can be carried out for more general surfaces:

Definition 3.6 ([Hub06]; cf. [Mar23])**.** A Riemann surface is of *finite type* if it is isomorphic to a compact surface from which at most finitely many points have been removed.

Since we will not deal with surfaces with boundary, we may imagine finite type surfaces as closed Riemann surfaces with finitely many punctures. It will be (sometimes) convenient for us to think of punctures as marked points. That is, instead of deleting the points, we can make them distinguished. Marked points and punctures carry the same topological information, so we can go back and forth between punctures and marked points as desired. Definitions 3.3, 3.4 and 3.5 can be extended to such surfaces by requiring that the set of punctures or marked points remain invariant (not necessarily pointwise). The notations change to $\mathcal{M}_{g,n}$, $\mathcal{T}_{g,n}$ and $\mathrm{Mod}_{g,n}$, respectively, where n denotes the number of marked points or punctures.

3.2 Measured foliations

Measured foliations were introduced by Thurston [Thu88] and provide the appropriate framework to describe geodesic flows on half-translation surfaces, as we will soon see. Before giving the general definition of a measured foliation, we discuss an example from [Far12] and restrict our attention to the case of the torus, where the situation is much simpler.

Let ℓ be any line through the origin in $\mathbb{R}^2$. The line ℓ determines a foliation $\tilde{\mathcal{F}}_\ell$ of $\mathbb{R}^2$ consisting of the set of all lines in $\mathbb{R}^2$ parallel to ℓ. Translations of $\mathbb{R}^2$ take lines to lines, and so any translation preserves $\tilde{\mathcal{F}}_\ell$ in the sense that it takes leaves to leaves.

Since all the deck transformations for the standard covering $\mathbb{R}^2 \to T^2$ are translations, the foliation $\tilde{\mathcal{F}}_\ell$ descends to a foliation $\mathcal{F}_\ell$ on T^2. If the slope of ℓ is rational, then every leaf of $\mathcal{F}_\ell$ is a simple closed geodesic in T^2. If the slope of ℓ is irrational then every leaf of $\mathcal{F}_\ell$ is a dense geodesic in T^2.

The foliations $\tilde{\mathcal{F}}_\ell$ come equipped with extra structure. Let $\nu_\ell : \mathbb{R}^2 \to \mathbb{R}$ be a map that records the distance from ℓ. Integration against the 1-form $\mathrm{d}\nu_\ell$ give a *transverse measure* on $\tilde{\mathcal{F}}_\ell$. What this means is that any smooth arc α transverse to the leaves of $\tilde{\mathcal{F}}_\ell$ can be assigned a length $\mu(\alpha) = \int_\alpha \mathrm{d}\nu_\ell$. The quantity $\mu(\alpha)$ is the total variation of α in the

direction perpendicular to ℓ. Thus, $\mu(\alpha)$ is invariant under isotopies of α that move each point of α within the leaf of $\tilde{\mathcal{F}}_\ell$ in which it is contained. The 1-form $\mathrm{d}\nu_\ell$ is preserved by translations, and so descends to a 1-form w_ℓ on T^2, and induces a transverse measure on the foliation $\mathcal{F}_\ell$. The structure of a foliation on T^2 together with a transverse measure is called a *transverse measured foliation* on T^2.

Note that a transverse measured foliation on T^2 is completely determined by the 1-form w_ℓ. The leaves of $\mathcal{F}_\ell$ in T^2 are simply the integral submanifolds to the distribution (see [Lee13, §19]) determined by the kernel of w_ℓ.

Now we want to transfer our discussion of measured foliations on the torus to closed surfaces of genus $g \geq 2$ (hyperbolic surfaces). The main obstruction is that one cannot have non-vanishing vector fields on such manifolds due to the following theorem:

Theorem 3.7 (Poincaré-Hopf; see [GP74, §3])**.** *Let M be a closed smooth manifold. Let X be a vector field on M with a set of isolated zeroes denoted by C. Then*

$$\sum_{p \in C} index_p(X) = \chi(M).$$

As a corollary we obtain that if there exists a non-vanishing vector field then $\chi(M) = 0$.

Definition 3.8. A *singular foliation* $\mathcal{F}$ on a closed surface S is a decomposition of S into a disjoint union of subsets of S called the *leaves* of $\mathcal{F}$, and a finite set of points S, called *singular points* of $\mathcal{F}$ such that the following two conditions hold.

(i) For each non-singular point $p \in S$ there is a smooth chart from a neighborhood of p to $\mathbb{R}^2$ that takes leaves to horizontal line segments. The transition maps between any two of these charts are smooth maps of the form $(x, y) \mapsto (f(x, y), g(y))$. In other words, the transition maps take horizontal lines to horizontal lines.

(ii) For each singular point $p \in S$ there is a smooth chart from a neighborhood of p to $\mathbb{R}^2$ that takes leaves to the level sets of a k-pronged saddle, $k \geq 3$ (see Figure 3.1).

Remark 3.2. This is not the consensus definition of a singular foliations, but serves well our purposes. Technically, let $\mathfrak{X}(S)$ be the space of vector fields on S, which is a module over the algebra $C^\infty(S)$. Then, a singular foliation $\mathcal{F}$ is an involutive and locally finitely generated $C^\infty(S)$-submodule of $\mathfrak{X}(S)$ (see [LLR22]).

From now on we will unambiguously use the term foliation for foliations that have singularities as well as for those that do not.

Let $\mathcal{F}$ be a foliation on a surface S. A smooth arc α in S is transverse to $\mathcal{F}$ if α misses the singular points of $\mathcal{F}$ and is transverse to each leaf of $\mathcal{F}$ at each point in its interior. Let $\alpha, \beta : I \to S$ be smooth arcs transverse to $\mathcal{F}$.

Definition 3.9. A *leaf-preserving isotopy* from α to β is a map $H : I \times I \to S$ such that

(i) $H(I \times \{0\}) = \alpha$ and $H(I \times \{1\}) = \beta$.

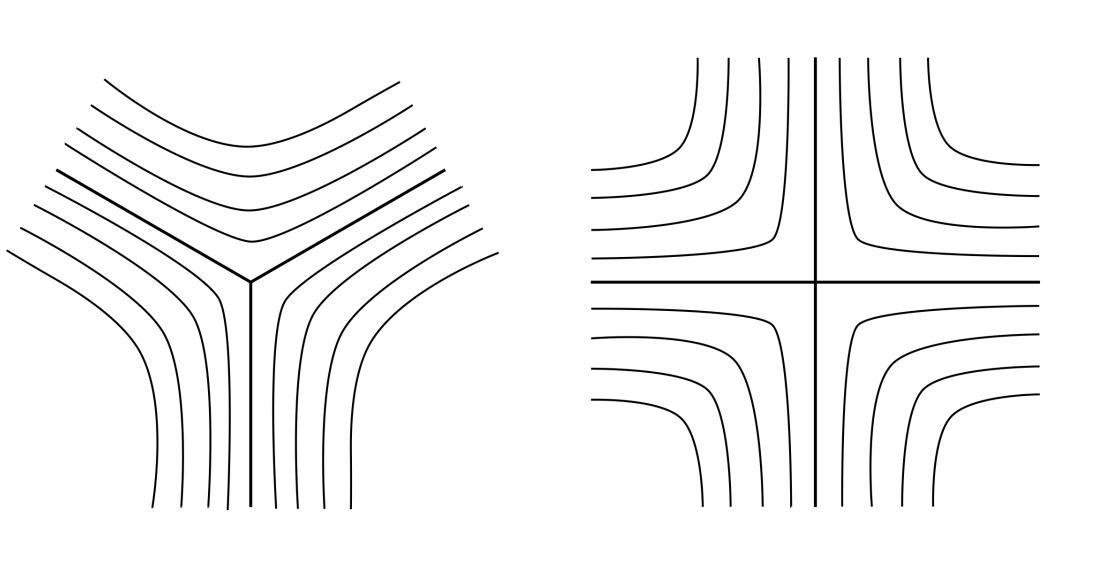


Figure 3.1: A 3-pronged saddle and a 4-pronged saddle at singular points of a measured foliation.

(ii) $H(I \times \{t\})$ is transverse to $\mathcal{F}$ for each $t \in [0, 1]$.

(iii) $H(\{0\} \times I)$ and $H(\{1\} \times I)$ are each contained in a single leaf.

Note that the second and third conditions imply that $H(\{s\} \times I)$ is contained in a single leaf for any $s \in [0, 1]$.

Definition 3.10. A *transverse measure* μ on a foliation $\mathcal{F}$ is a function that assigns a positive real number to each smooth arc transverse to $\mathcal{F}$, so that μ is invariant under leaf-preserving isotopy, and μ is regular (i.e. absolutely continuous) with respect to Lebesgue measure.

In other words, this last condition means that each point of S has a neighborhood U and a smooth chart $U \to \mathbb{R}^2$ so that the measure μ is induced by $|dy|$ on $\mathbb{R}^2$.

Definition 3.11. A *measured foliation* $(\mathcal{F}, \mu)$ on a surface S is a foliation $\mathcal{F}$ of S equipped with a transverse measure μ.

We say two measured foliations are transverse if their leaves are transverse away from the singularities. Note that transverse measured foliations must have the same set of singularities.

Definition 3.12. A measured foliation $(\mathcal{F}, \mu)$ on a surface S is said to be *orientable* if there exists a continuous, nowhere-zero vector field on $S \setminus \Sigma$ that is everywhere tangent to the leaves of $\mathcal{F}$, where $\Sigma \subset S$ denotes the set of singular points of $\mathcal{F}$.

Equivalently, a measured foliation is orientable if it arises as the foliation defined by the level sets of a globally defined closed 1-form. In this case, the orientation on each leaf is induced by the direction of the 1-form, and the orientation is consistent throughout the surface. If such a global 1-form does not exist, the foliation is said to be *non-orientable*. An orientable foliation induces a flow along the leaves, which is called uniquely ergodic if the transverse measure μ is unique up to scalar multiplication, as in Definition 1.30.

Construction of a measured foliation from a polygon

Given any closed surface S, it is well known that we can realize S as the quotient of a polygon P in $\mathbb{R}^2$ by side identifications. We impose two additional conditions:

(i) Any time two edges of P are identified, they are parallel.

(ii) The total Euclidean angle around each point of S is greater than π (this only needs to be checked at the vertices of P).

Any foliation of $\mathbb{R}^2$ by parallel lines restricts to give a foliation of the (interior of) P. We claim that this foliation induces a foliation of S. It is easy to see that any point of S coming from a point of P that is not a vertex of P has a regular neighborhood that satisfies the definition of a regular point of a foliation.

So what happens at a point $p \in S$ corresponding to a vertex of P? The first observation is that, since identified sides of P are parallel, the total angle around P is an integer multiple of π. In particular, there is some vertex $\tilde{p}$ of P in the preimage of p, and a vector v based at $\tilde{p}$ that points into P (possibly along an edge) and is parallel to the foliation of P. If we sweep out an angle of π starting with v, we find a closed Euclidean half-disk in S that is foliated by lines parallel to the diameter. If we continue to sweep out angles of π, we see that a neighborhood of p looks like some number of Euclidean half-disks, each foliated by lines parallel to the diameter, and glued along oriented radii. By our assumption on the total angle around each point of S coming from a vertex of P, we know that there are at least two half-disks glued at p. If there are exactly two half-disks, then p is a regular point. If there are k half-disks, where $k \geq 3$, then p is a singularity with k prongs.

One measure on the induced foliation of S is the one given by the total variation of the Euclidean distance in the direction perpendicular tot the foliation of P. The charts we described above are the natural charts for the non-singular points.

Suppose that, in this construction, we orient each edge of P so that the identifications respect these orientations. If all side pairings identify sides of P that are parallel in the oriented sense (as opposed to anti-parallel), then the resulting foliation of S is orientable. Indeed, either of the two orientations of the foliation on the interior of P extend to give orientation of the entire foliation of S.

3.3 Quadratic differentials

The (holomorphic) cotangent bundle or *canonical bundle* $T^*M \to M$ to a Riemann surface M is the complex line bundle over M whose fiber above a point p is the space of complex linear maps $T_pM \to \mathbb{C}$. A holomorphic 1-form is a holomorphic section of the cotangent bundle of M.

Definition 3.13 ([Far12, §11.3.1])**.** A *holomorphic quadratic differential* on M is a holomorphic section os the symmetric square of the cotangent bundle of M, i.e., a quadratic form on the tangent bundle that varies holomorphically. It is sometimes denoted by $H^0(M, \Omega^{\otimes 2})$, where Ω denotes the canonical bundle on M.

Alternatively, as in the classic text [Str84], we can describe a holomorphic quadratic differential on M in terms of local coordinates. Let $\{z_\alpha : U_\alpha \to \mathbb{C}\}_{\alpha \in A}$ be an atlas for M. A holomorphic quadratic differential q on M is specified by a collection of expressions $\{\phi_\alpha(z_\alpha)\mathrm{d}z_\alpha^2\}_{\alpha \in A}$ (where $\mathrm{d}z^2 = \mathrm{d}z \otimes \mathrm{d}z$) with the following properties:

(i) Each $\phi_\alpha : z_\alpha(U_\alpha) \to \mathbb{C}$ is a holomorphic function with a finite set of zeroes.

(ii) For any two coordinates charts z_α and z_β, we have

$$\phi_\beta(z_\beta)\left(\frac{\mathrm{d}z_\beta}{\mathrm{d}z_\alpha}\right)^2 = \phi_\alpha(z_\alpha). \tag{3.3}$$

Condition (ii) may be stated as: the collection $\{\phi_\alpha(z_\alpha)\mathrm{d}z_\alpha^2\}_{\alpha \in A}$ is invariant under change of local coordinates. To say this in another way: if q is given in one chart by $\phi_\alpha(z)$, and in another chart by $\phi_\beta(z)$, and the change of coordinates from the first chart to the second is the holomorphic map ψ, then

$$\phi_\beta(z)(\mathrm{d}\psi)^2 = \phi_\alpha(z).$$

It follows that the order of a zero of a holomorphic quadratic differential is well-defined, independent of the chart. We call the zeroes (poles are not allowed) of q its *critical points* and the points where q does not vanish its *regular points*.

3.3.1 Natural coordinates

Definition 3.14. Let M be a Riemann surface. A *natural coordinate* (U, ζ) for a quadratic differential q on M is a holomorphic chart such that $q(z) = \zeta^k \mathrm{d}\zeta^2$, for some $k \geq 0$.

Such coordinates exists. We follow [Str84]. On the one hand, let $p \in M$ be a regular point of q and (U, z) a chart such that $z(p) = 0$ and U is small enough such that it does not contain any zeroes of q. In this coordinates suppose that $q(z) = \phi(z)\mathrm{d}z^2$. Our goal is to show that there is a local coordinate ζ at p so that $q(\zeta) = \mathrm{d}\zeta^2$. Let $\sqrt{q} = \sqrt{\phi(z)}\mathrm{d}z$ be a holomorphic 1-form. Let

$$\zeta = \Phi(z) = \int_0^z \sqrt{q} = \int_0^z \sqrt{\phi(w)}\mathrm{d}w,$$

where some branch os the square root function is chosen (this is possible since $\phi \neq 0$). Of course, when we integrate from 0 to z, we mean to integrate along some (any) path from 0 to z inside a sufficiently small simply connected neighborhood of 0. We have $\mathrm{d}\zeta = \Phi'(z)\mathrm{d}z = \sqrt{\phi(z)}\mathrm{d}z$. Now, applying (3.3) it is clear that in the ζ coordinate the representation of q is

$$q(\zeta) = \mathrm{d}\zeta^2,$$

thus ζ is natural.

On the other hand, let $p \in M$ be a critical point q of order $k \geq 1$. There are two cases depending on the parity of k, but they yield the same type of coordinates, so we will only present the most delicate one.

Let $k \geq 1$ be odd. In a local coordinate z near p with $z(p) = 0$, we have $q(z) = z^k h(z)\mathrm{d}z^2$, where $h(0) \neq 0$. Since $h(z)$ is holomorphic, we can expand it as $h(z) = a_0 + a_1 z + \cdots$ with $a_0 \neq 0$. In a sufficiently small neighborhood of zero we can select a single valued branch of the square root of $h(z) = a_0 + a_1 z + \cdots$, say $\sqrt{h(z)} = b_0 + b_1 z + \cdots$. We then get, locally outside $z = 0$, the 1-form $\sqrt{q} = z^{k/2}\sqrt{h(z)}\mathrm{d}z = z^{k/2}(b_0 + b_1 z + \cdots)\mathrm{d}z$. We define

$$\eta = \Phi(z) = \int_0^z \sqrt{q} = \int_0^z w^{k/2}\sqrt{h(w)}\mathrm{d}w = \int_0^z w^{k/2}(b_0 + b_1 w + \cdots)\mathrm{d}w,$$

and by integrating term by term we obtain

$$\eta = \Phi(z) = z^{\frac{k+2}{2}}(c_0 + c_1 z + \cdots), \quad \text{with } c_j = \frac{2b_j}{k + 2(j+1)}.$$

Let $d_0 + d_1 z + \cdots = (c_0 + c_1 z + \cdots)^{\frac{2}{k+2}}$ be a single valued branch of the right hand side in some sufficiently small neighborhood of $z = 0$. Then,

$$\eta = \Phi(z) = \zeta^{\frac{k+2}{2}}, \quad \text{where } \zeta = z(d_0 + d_1 z + \cdots),$$

which is a conformal homeomorphism of some neighborhood of $z = 0$ onto a disk $|\zeta| < \rho$.

The representation of the quadratic differential q in terms of the parameter ζ is found as follows:

$$\mathrm{d}\eta = \Phi'(\zeta)\mathrm{d}\zeta = \frac{k+2}{2}\zeta^{k/2}\mathrm{d}\zeta,$$

and therefore

$$\mathrm{d}\eta^2 = \left(\frac{k+2}{2}\right)^2 \zeta^k \mathrm{d}\zeta^2$$

which is again single valued in the full neighborhood. Therefore, q has a local representation in the natural ζ coordinate

$$q(\zeta) = \left(\frac{k+2}{2}\right)^2 \zeta^k \mathrm{d}\zeta^2,$$

which can be easily rescaled to satisfy Definition 3.14.

3.3.2 The q-metric

Natural coordinates for a holomorphic quadratic differentials q on a Riemann surface M endow it with a *singular Euclidean metric* which we will refer to as the q-metric, with length element given by

$$|\mathrm{d}s| = |\phi(z)|^{1/2}|\mathrm{d}z| = |\phi(z)|^{1/2}\sqrt{\mathrm{d}x^2 + \mathrm{d}y^2}.$$

Let us see what this means precisely. Around a regular point for q, choose a natural coordinate ζ, so that $q = \mathrm{d}\zeta^2$. Then it is clear that $|\mathrm{d}s| = |\mathrm{d}\zeta|$, the standard Euclidean metric in $\mathbb{C}$. Thus, a quadratic differential induces a flat metric on M away from the critical points of q. In contrast, choose a zero of q of order $k \geq 1$ and a natural coordinate ζ so that $q = \zeta^k \mathrm{d}\zeta^2$. The metric is given by $|\mathrm{d}s| = |\zeta|^{k/2}|\mathrm{d}\zeta|$. Change to polar coordinates $\zeta = re^{i\theta}$, thus $|\zeta| = r$ and $\mathrm{d}\zeta = e^{i\theta}(\mathrm{d}r + ir\mathrm{d}\theta)$. Withing this setting we have $|\mathrm{d}\zeta| = (\mathrm{d}r^2 + r^2\mathrm{d}\theta^2)^{1/2}$, hence

$$|\mathrm{d}s| = r^{k/2}\sqrt{\mathrm{d}r^2 + r^2\mathrm{d}\theta^2} = \sqrt{r^k\mathrm{d}r^2 + r^{k+2}\mathrm{d}\theta^2}$$

Redefine the radial coordinate

$$\rho := \frac{2}{k+2}r^{(k+2)/2}, \quad \text{inverting } r = \left(\frac{k+2}{2}\rho\right)^{2/(k+2)}.$$

Taking derivatives we have that $\mathrm{d}\rho = r^{k/2}\mathrm{d}r$. Hence

$$\begin{aligned} r^k\mathrm{d}r^2 &= (r^{k/2}\mathrm{d}r)^2 = \mathrm{d}\rho^2, \\ r^{k+2}\mathrm{d}\theta^2 &= \left(r^{(k+2)/2}\right)^2\mathrm{d}\theta^2 = \left(\frac{k+2}{2}\rho\right)^2\mathrm{d}\theta^2. \end{aligned} \tag{3.4}$$

Substituting back from equations (3.4) we obtain the length element

$$|\mathrm{d}s| = \sqrt{\mathrm{d}\rho^2 + c\rho\mathrm{d}\theta^2}, \quad \text{where } c = \frac{k+2}{2}. \tag{3.5}$$

Equation (3.5) corresponds to a metric with a cone singularity of angle $2\pi c = (k+2)\pi$. Intuitively, this implies that if you walk around a zero of q of order k the angle you transverse is not 2π like in flat Euclidean space, but a greater span because $k \geq 1$ (see Figure 3.2). This was expected, since we are working with hyperbolic surfaces ($g \geq 2$) and Gauss-Bonnet theorem forces the curvature to concentrate on finitely many singular points.

We can speak of parallel translation with respect to the connection determined by the q-metric. The global obstruction to parallel translation being well-defined is the holonomy group which is a subgroup of $O(2)$. If a metric of zero curvature arises from a quadratic differential, then the holonomy group is either the trivial group or $\{I, -I\}$ (see [KMS86]). One says the quadratic differential is *orientable* in the first case and *non-orientable* in the second case. Note that the underlying surface is always orientable. Conversely, if a surface of zero curvature has cone type singularities and the holonomy group is one of these two groups, the the metric comes from a quadratic differential.

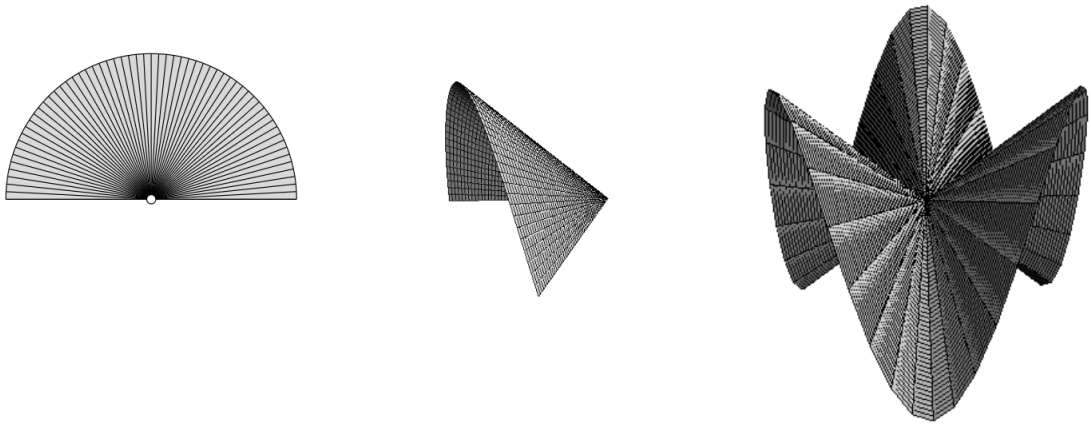


Figure 3.2: A neighborhood of a conical point with a cone angle 6π can be glued from six metric half discs ([Zor06]).

For our purposes, we will only be interested in orientable quadratic differentials, which can be equivalently defined as global squares $q = \omega^2$ of holomorphic abelian differentials (1-forms). The reason to doing so is two-fold: firstly, quadratic differentials arising from rational billiards (see §4.1) are always orientable, and secondly, usually (but not always) some results about abelian differentials can be extended to the non-orientable setting by a double cover construction, that is, one consider a (canonical) double cover of the initial Riemann surface, equipped with a non-orientable quadratic differential q, such that a global square of the lift of q is well-defined.

The q-metric also possesses a local area form given by

$$\mathrm{d}A = \frac{1}{2i}|\phi(z)|\mathrm{d}\bar{z} \wedge \mathrm{d}z = |\phi(z)|\mathrm{d}x \wedge \mathrm{d}y,$$

which allows us to define an L^1-norm

$$\|q\| = \int_M |q| = \int_M |\phi(z)|\mathrm{d}x \wedge \mathrm{d}y$$

and call it, with good reasons, the *area* of q. This norm clearly induces a topology and a (Finsler) metric on the space of holomorphic quadratic differentials on M modulo the equivalence relation of being isomorphic by means of a homeomorphism isotopic to the identity. This space is denoted by Q_g and can be identified with the cotangent bundle of $\mathcal{T}_g$ (via duality with Beltrami differentials) and given the bundle topology, which coincides with the L^1-norm defined above.

The trajectory structure in terms of geodesics of the q-metric is essentially that of a measured foliation (see [Str84]). For regular points on a sufficiently small neighborhood it consists of straight lines in the ζ-plane and for critical points of order $k \geq 1$ we obtain $(k+2)$-pronged saddles (see Figure 3.3).

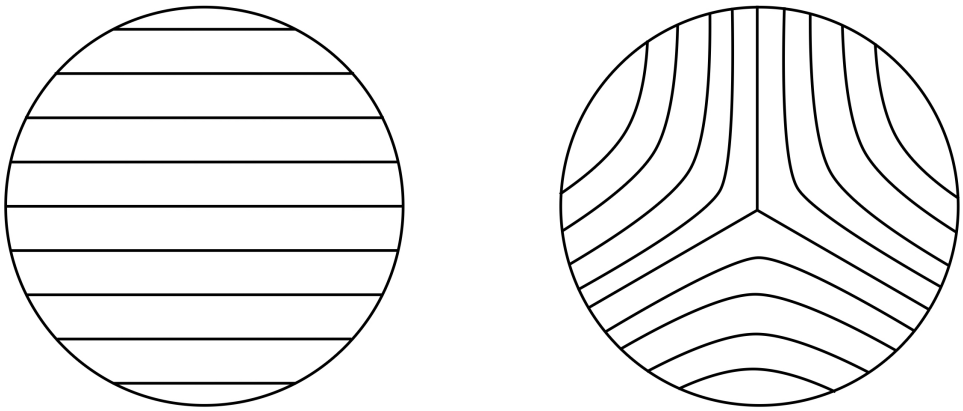


Figure 3.3: Local trajectories of a quadratic differential: near a regular point and near a zero of order 1.

In particular, given a quadratic differential q on a Riemann surface M, we obtain a measured foliation by taking the union of the zeroes of q with the set of smooth paths in M whose tangent vectors evaluate to positive real numbers under q. This foliation is called the *horizontal foliation* for q. If we instead take the paths in M whose tangent vectors evaluate to negative real numbers under q, the resulting foliation is called the *vertical foliation* for q. Clearly, if q is orientable the horizontal and vertical measured foliations are orientable.

Say that, in some chart a holomorphic quadratic differential is given by the expression $\phi(z)\mathrm{d}z^2$. In any given chart, the functions

$$\mu^h(\alpha) = \int_\alpha \left|\mathrm{Im}\left(\sqrt{\phi(z)}\mathrm{d}z\right)\right|, \quad \mu^v(\alpha) = \int_\alpha \left|\mathrm{Re}\left(\sqrt{\phi(z)}\mathrm{d}z\right)\right|$$

induce transverse measures μ_q^h and μ_q^v on the horizontal and vertical foliations of q, respectively. Therefore, every quadratic differential on a Riemann surface induces a pair of transverse measured foliations. Naturally, one can ask if all measured foliations on M arise as the horizontal (or vertical) foliations of a quadratic differential, and the answer, due to Hubbard and Masur [HM79], is positive up to a certain equivalence.

In natural coordinates, the horizontal and vertical foliations arising from the quadratic differential q (locally) correspond to the horizontal and vertical foliations of the complex plane. Indeed, we are going to prove that the horizontal foliation corresponds to $\mathrm{Im}\sqrt{q} = 0$ (the vertical case is analogous). Indeed, in a natural coordinate ζ away from the zeroes of q we have $q = \mathrm{d}\zeta^2$. Let us write $\zeta = x + iy$ so $\mathrm{d}\zeta = \mathrm{d}x + i\mathrm{d}y$. Let v be a real tangent vector at a point, expressed in local coordinates as $v = v_x\,\partial_x + v_y\,\partial_y$. Then

$$\mathrm{d}\zeta(v) = \mathrm{d}x(v) + i\,\mathrm{d}y(v) = v_x + iv_y.$$

Let us denote $a := v_x$ and $b := v_y$ so that

$$\sqrt{q}(v) = \mathrm{d}\zeta(v) = a + ib.$$

Compute

$$q(v) = (\mathrm{d}\zeta(v))^2 = (a+ib)^2 = a^2 - b^2 + 2iab, \tag{3.6}$$

and therefore $\mathrm{Im}\big(q(v)\big) = 2ab$. In particular, $q(v) \in \mathbb{R}$ if and only if $ab = 0$. From (3.6) we have

$$\mathrm{Im}\left(\sqrt{q}(v)\right) = \mathrm{Im}(a+ib) = b.$$

Suppose $q(v) > 0$. Then $q(v) \in \mathbb{R}$, so $ab = 0$. But $q(v) = a^2 - b^2 > 0$, so $a \neq 0$. Hence $b = 0$, and so $\mathrm{Im}\left(\sqrt{q}(v)\right) = b = 0$. Conversely, suppose $\mathrm{Im}\left(\sqrt{q}(v)\right) = 0$, so $b = 0$. Then

$$q(v) = (a+ib)^2 = a^2 > 0.$$

This proves the equivalence. Moreover, in this setting the transverse measure is $|\mathrm{d}y|$ (resp. $|\mathrm{d}x|$), that is, the net vertical (resp. horizontal) variation of a transverse path.

3.3.3 Half-translation structures and the $\mathbf{SL}(2,\mathbb{R})$ action

In order to introduce the action of $\mathrm{SL}(2,\mathbb{R})$ on q we need a more geometric approach. Let S be a compact orientable topological surface of genus $g \geq 1$. We start from the following combinatorial data:

- A non-emtpy finite subset $\Sigma = \{p_1, \ldots, p_s\} \subset S$.
- An associated family $\kappa = (\kappa_1, \ldots, \kappa_s)$ of positive integers which should be seen as *ramification indices*. These must satisfy certain restrictions related to g, but we will not need them.

Definition 3.15 (cf. [Yoc10])**.** A *half-translation structure* on (S, Σ) is a maximal atlas ζ for $S \setminus \Sigma$ by open sets of $\mathbb{C} \cong \mathbb{R}^2$ which satisfies the two following properties:

(i) Any coordinate change between two charts of the atlas is locally of the form $z \mapsto \pm z + c$, with $c \in \mathbb{C}$.

(ii) For every $1 \leq i \leq s$, there exists a neighborhood V_i of p_i, a neighborhood W_i of 0 in $\mathbb{R}^2$ and a ramified covering $\pi : (V_i, p_i) \to (W_i, 0)$ of degree κ_i such that every injective restriction of π is a chart of ζ.

A half-translation structure naturally induces a complex structure on $S \setminus \Sigma$, which can be uniquely extended to the whole S by Riemann's theorem on removable singularities. If condition (i) is restricted to transitions of the form $z \mapsto z + c$ then the structure is called a *translation surface*.

One of the most significant features of the choice of a holomorphic quadratic differential q on a Riemann surface M is that it induces this so-called half-translation structure. Indeed, they are essentially the same object. Kerchoff, Masur and Smillie [KMS85] even state

Geometric language	**Complex-analytic language**
Half-translation structure	Complex structure + a choice of a holomorphic quadratic differential
Conical point with cone angle $(k+2)\pi$	Zero of degree k of q (in local coordinates $q = \zeta^k \mathrm{d}\zeta^2$)
Area of the half-translation surface S	$\|q\| = \int_S \|q\|$
Measured foliation	Vertical and horizontal trajectories of q

Table 3.1: Correspondence between geometric and complex-analytic descriptions.

> We define a quadratic differential on a surface to be a structure determined by an atlas of charts mapping open subsets of M to $\mathbb{R}^2$. We require that the change-of-coordinate functions be restrictions of maps of the form $v \to (v+c)$ or $v \to (-v+c)$. We also allow a finite number of singular points at which the charts are n-fold branched covering of $\mathbb{R}^2/(\pm 1)$.

Consider a pair (M, q) of a Riemann surface M with a choice of a holomorphic quadratic differential with critical points (zeroes) $\Sigma = \{p_1, \dots, p_s\} \subset S$. Away from the critical set consider natural parameters ζ_1, ζ_2 whose domains overlap. In the overlap, $\zeta_1 = \phi(\zeta_2)$ for some holomorphic function ϕ. By eq. (3.3) we have $(\phi'(\zeta_2))^2 = 1$, so condition (i) is satisfied. Condition (ii) comes from natural coordinates around critical points.

Conversely, consider a half-translation structure (S, Σ) on the underlying topological surface of M with singularity type $\kappa = (\kappa_1, \dots, \kappa_s)$. In each coordinate chart (U_α, ϕ_α) of $S \setminus \Sigma$ we set

$$q|_{U_\alpha} := \phi_\alpha^*(\mathrm{d}z^2).$$

Suppose $U_\alpha \cap U_\beta \neq \varnothing$ and rename $\phi_\alpha = z_\alpha$ and $\phi_\beta = z_\beta$. Then on the overlap $z_\beta = \pm z_\alpha + c$, so $\mathrm{d}z_\beta = \pm \mathrm{d}z_\alpha$. Hence $\mathrm{d}z_\beta^2 = \mathrm{d}z_\alpha^2$ and $\phi_\alpha^*(\mathrm{d}z^2) = \phi_\beta^*(\mathrm{d}z^2)$.

Now let $p_i \in \Sigma$, then in a small disk around p_i we can write $\pi_i(z) = z^{\kappa_i}$. We pull back again $\mathrm{d}z^2$ via this covering, $q = \pi_i^*(\mathrm{d}z^2)$. More explicitly,

$$q = \pi_i^*(\mathrm{d}z^2) = (d\pi_i)^2 = \kappa_i^2 z^{2\kappa_i - 2} \mathrm{d}z^2,$$

which has a zero of order $2\kappa_i - 2$ at p_i. All in all, we conclude that q is a global holomorphic section of $\Omega^{\otimes 2}$.

We explain two actions on Q_g. First, let $Q_g^0 = \{q \in Q_g : \|q\| = 1\}$. There is an action $\mathrm{Mod}_g \times Q_g \to Q_g$ of the mapping class group analogous to the action on the space of measured foliations described in [Mas85], namely, by pullback: $[\varphi] \cdot q = (\varphi^{-1})^* q$. This allows us to define the *moduli space of unit norm quadratic differentials*

$$\mathcal{Q}_g = Q_g^0 / \mathrm{Mod}_g.$$

Secondly, let $\{\phi_i\}$ be an atlas of q and let $a \in \mathrm{SL}(2,\mathbb{R})$. We can define a new atlas $\{a\phi_i\}$. In this way we define an action of $\mathrm{SL}(2,\mathbb{R}) \curvearrowright Q_g^0$, which is at the heart of the main result of Chapter 4. Note that if $a = -I$ then the two atlases give isomorphic structures. Thus this action descends to a well-defined action of $\mathrm{PSL}(2,\mathbb{R}) = \mathrm{SL}(2,\mathbb{R})/\{I,-I\}$. The notion of parallel translation remains unchanged when the coordinate charts are changed as above. In particular the set of geodesic curves remains unchanged.

Finally, having already devised the multifaceted nature of quadratic differentials, we include in Table 3.1 a correspondence between geometric and complex-analytic notions (in the spirit of [Zor06]).

Chapter 4

Unique ergodicity of billiard flows

In this chapter we prove that the billiard flow is uniquely ergodic within the invariant surface M_θ given by the Zemlyakov-Katok unfolding, for almost every $\theta \in S^1$. This is a direct consequence of

Theorem 4.1 ([KMS86])**.** *Given a compact Riemann surface M and a holomorphic quadratic differential q, then for almost every θ the vertical foliation of $e^{i\theta}q$ is uniquely ergodic.*

This more general theorem follows from Theorem 4.5 and Theorem 4.22, which we prove in §4.2 and §4.22, respectively. In §4.1 we explain why Theorem 4.1 is equivalent to the unique ergodicity of the billiard flow in almost every direction with the references [ZK75],[KMS86] and [MT02].

The following one parameter subgroups of $\mathrm{PSL}(2,\mathbb{R})$ will be important in what follows:

$$g_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix},$$
$$r_\theta = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix},$$
$$h_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}.$$

We refer to g as the geodesic or *Teichmüller flow*. This can be thought of as the geodesic flow on the moduli space. The action of r_θ on a quadratic differential q is the same as multiplying q by $e^{2i\theta}$. Since $r_{\theta+\pi} = -r_\pi$, multiplication by r_θ defines an action of the circle $\mathbb{R}/\pi\mathbb{Z}$ on the set of quadratic differentials. Note that the action of r_θ leaves the flat metric invariant. Any two quadratic differentials which determine the same metric and orientation on M differ by the action of this subgroup. We refer to h as the *horocycle flow*.

The groups g and r act continuously on $\mathcal{Q}_g$. Since $\mathrm{PSL}(2,\mathbb{R})$ is generated by these subgroups it acts continuously on $\mathcal{Q}_g$.

4.1 Zemlyakov-Katok unfolding

Let us come back momentarily to rational billiards. This section deals with how we reduce the problem of billiard dynamics to a problem concerning the vertical flow of the measured foliation induced by a quadratic differential.

Let $\mathcal{P}$ be a rational billiard of n sides, G its associated group of reflections and $(\tilde{\Omega}, \Phi)$ the billiard flow for the points with infinite trajectories. Since $\mathcal{P}$ is a rational polygon, we proved that G is finite and so is the action $G \curvearrowright S^1$. To analyze the billiard flow modulo these symmetries, we choose a fundamental domain $J \subset S^1$, an open arc containing one representative from each orbit under the action of G. We can define a function from $\mathcal{P} \times S^1$ to J by composing the projection onto S^1 with the quotient map from S^1 to J. This composition is compatible with the identifications defining $\tilde{\Omega}$ and therefore induces a function $\pi : \tilde{\Omega} \to J$. Note that π is constant on orbits of Φ.

Let θ be an interior point of the interval J. There is a (punctured) surface $M_\theta = \pi^{-1}(\theta)$. Since each such surface is invariant it makes sense to analyze the flow restricted to any such surface. Each surface inherits from $\mathcal{P}$ a flat metric with cone type singularities. The flow $\Phi|_{M_\theta}$ is generated by a vector field on M_θ which is parallel with respect to the flat structure on M_θ. With respect to their flat metrics the surfaces M_θ are all isometric. We will describe a natural identification of any two of them. Let θ and λ be points in the interior of J. If (x, v) is in M_θ then we identify it with (x, v'), where v' corresponds to λ under the quotient identification and v' belongs to the same fundamental domains as v under the action of G on S^1. With this identification M_θ, when thought of as a Euclidean manifold, is independent of θ. We will denote it by M. We can choose a direction on M and thus a quadratic differential q so that each of the flows $\Phi|_{M_\theta}$ is equivalent to one of the flows generated by some $r_\theta q$ for $\theta \in J$, that is, by its vertical foliation.

This can be seen not so technically but more intuitively. We explain the equivalent point of view of Zemlyakov and Katok [ZK75], although some papers atribute it to Fox and Kershner [FK36]. The Zemlyakov-Katok unfolding consists of reflecting the billiard at each collision with a side and continuing the trajectory in a straight line. Bear in mind that we only need $|G|$ reflections of $\mathcal{P}$ to keep track of all possible directions of $\mathrm{Orb}_G(\theta)$ from a prefixed non-exceptional direction θ. Identifying the sides appropriately after such reflections we obtain a compact orientable surface which does not depend on θ (see Figure 4.1). In particular,

Proposition 4.2 ([MT02])**.** *Let the angles of $\mathcal{P}$ be $\pi m_i/n_i$, $i = 1, \ldots, k$ where $\gcd(m_i, n_i) = 1$ for every i, and let N be the least common multiple of the n_i's. Then the genus of M is*

$$g = 1 + \frac{N}{2}\left(k - 2 - \sum \frac{1}{n_i}\right)$$

Proof. We need to analyse how the gluings are made around a vertex of $\mathcal{P}$. Consider the i-th vertex V with angle $\pi m_i/n_i$. Let G_i be the group of linear transformations in the plane generated by the reflections in the sides of $\mathcal{P}$ adjacent to V. Then G_i consists of $2n_i$ elements.

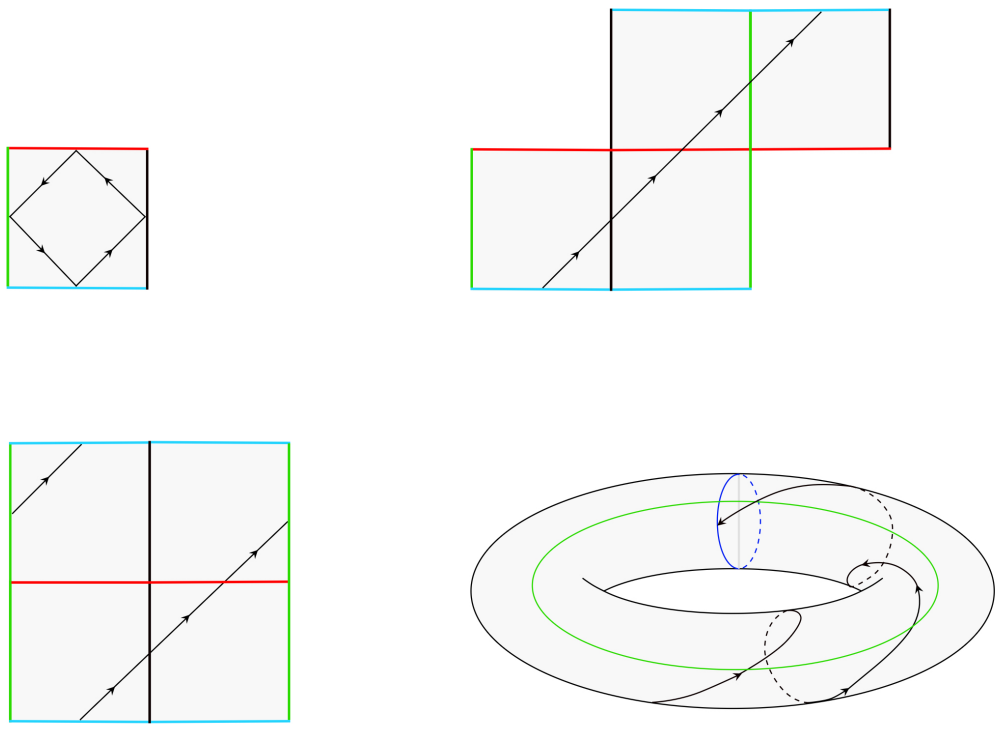


Figure 4.1: Unfolding of a trajectory in a square to a geodesic on a torus.

According to the construction of M the number of copies of the polygons $\mathcal{P}_j$ that are glued together at V equals the cardinality of the orbit of the test angle θ under the group G_i, that is $2n_i$. Originally we had $2N$ copies of $\mathcal{P}$, and therefore, $2N$ copies of the vertex V; after the gluings we have N/n_i copies of the vertex on M.

It follows that the total number of vertices in M is $\sum 1/n_i$. The total number of edges is Nk, and the number of faces is $2N$. Therefore the Euler characteristic of M equals

$$N \sum \frac{1}{n_i} - Nk + 2N = 2 - 2g,$$

where g is the genus, and the result follows. □

The key in relation to quadratic differentials is that M carries naturally a translation structure with singular points at the (image under the quotient of the) vertices (for instance, see [Mas22]). In light of what we have seen in §3, M comes equipped with a choice of a quadratic differential q, which is orientable since it is the square of the holomorphic one form pulled back from $\mathrm{d}z$ via the charts of the translation structure. As we have seen, quadratic differentials induce measured foliations, namely, a horizontal and a vertical foliation. Since

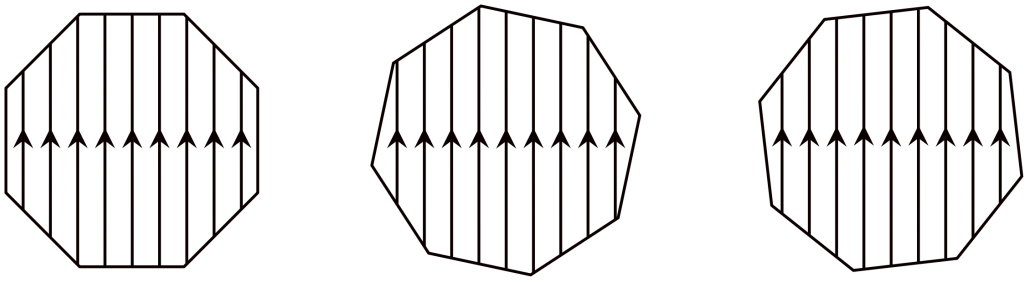


Figure 4.2: Vertical foliation of a quadratic diffential under the action of r_θ.

q is orientable, these foliations are also orientable, so we can define a flow that preserves the transverse measure. Finally, we ask ourselves if this flow is uniquely ergodic.

Another very important observation is that we should think of the quadratic differential q (the one arising from the unfolding) as its underlying unfolded polygon produced by $\mathcal{P}$ without identifications. This is, clearly, only for intuitive perception. Then, $r_\theta q$ should be interpreted as rotating this unfolded polygon. Therefore, studying the vertical foliation of $r_\theta q$ for every $\theta \in S^1$ is the same as studying the geodesic flow on M in every direction (see Figure 4.2). However, beware that not every quadratic differential arises as the unfolding of a rational billiard, so even though we are going to think about them as such, actually Theorem 4.1 is stronger than just proving unique ergodicity of billiard flows in a fixed direction.

4.2 The Teichmüller flow of $r_\theta(q)$

Definition 4.3. A *geodesic segment* is a geodesic interval with singularities as endpoints which contains no singularities in its interior.

Recall that the trajectory structure of a quadratic differential around regular points are straight lines. In fact, if a geodesic is contained in a coordinate chart, its image in $\mathbb{C} \cong \mathbb{R}^2$ is a line segment (also see [Str84]). We define the *horizontal* and *vertical components* of the curve to be the horizontal and vertical components of its image. If a geodesic path or curve is not contained in a single chart then we define its horizontal and vertical components by dividing it into short pieces and summing the components of the pieces. Two isotopic geodesic curves have the same horizontal and vertical components.

Action of $\mathrm{PSL}(2,\mathbb{R})$ on geodesic segments: let A be a matrix representing an element of $\mathrm{PSL}(2,\mathbb{R})$. Let β be a geodesic segment. Let h_1 and v_1 the horizontal and vertical

component of β. Let h_2 and v_2 be the coordinates of β with respect to Aq. Then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} h_1 \\ v_1 \end{pmatrix} = \begin{pmatrix} h_2 \\ v_2 \end{pmatrix}.$$

Note that in this equation the matrix and two vectors are well defined modulo multiplication by -1.

Definition 4.4. A quadratic differential q is *divergent* if $g_t(q)$ eventually leaves all compact sets as $t \to \infty$. A quadratic differential is *recurrent* if it is not divergent.

Note that q is recurrent if and only if the ω-limit set (Definition 1.25) of q is non-empty, i.e. if $\omega(q) = \{q' \in \mathcal{Q} : \exists t_n \to \infty \text{ with } g_{t_n}(q) \to q'\} \neq \varnothing$.

The following formulas will be used in the proof of the next theorem:

$$g_t h_{s\cdot\exp(-t)} g_{-t} = h_s, \tag{4.1}$$

$$\lim_{t\to\infty} g_t r_{s\cdot\exp(-t)} g_{-t} = h_s. \tag{4.2}$$

4.2.1 Case I: r_θ recurrent

Theorem 4.5. *The set of θ for which $r_\theta(q)$ is recurrent and not uniquely ergodic has measure zero.*

Proof. According to [Mas82] there is a subset B of quadratic differential space such that if q is not uniquely ergodic then every ω-limit point of the Teichmüller orbit of q is contained in B. For our purposes it suffices to note two properties of B. First, B is a closed set and second, B consists of quadratic differentials each of which contains a closed vertical geodesic (which may contain singularities). The second property implies that the intersection of B with any horocycle orbit is countable. To see this, let q be a quadratic differential. Let α be a closed geodesic with components h and v. The horizontal component of α with respect to $h_s(q)$ is $h + sv$. Also α is vertical when $h + sv = 0$. This occurs for at most one value of s. If β is isotopic to α then it will have the same components and be vertical for the same s. The number of isotopy classes of geodesics is countable; thus, the number of values s for which $h_s(q)$ can have a vertical geodesic is countable.

Assume that the theorem is false. Let U be a positive measure set of θ so that for θ in U, $r_\theta(q)$ is not uniquely ergodic and the orbit of $r_\theta(q)$ under the Teichmüller flow, g_t, has non-trivial ω-limit set. Partition quadratic differential space into a countable collection of compact sets. For each of theses sets K consider the set of directions θ for which the ω-limit set of $r_\theta(q)$ intersects K. Now U is the union of this countable collection of sets; thus, at least of these sets must have positive measure. Call this set V and the corresponding compact set K.

Let B_ϵ be the closed ϵ neighborhood of B.

Lemma 4.6. *For every δ there is an ϵ such that $\mu\{s \in [-1,1] : h_s(k) \in B_\epsilon\} < \delta$ for each $k \in K$.*

. Let $f_n(k) = \mu\{s \in [-1,1] : h_s(k) \in B_{1/n}\}$. The lemma is equivalent to the assertion that the supremum of f_n goes to 0 as n increases. This is a consequence of the following two properties of a sequence of functions defined on a compact set: f_n is a non-increasing sequence of functions with pointwise limit 0. Each function f_n is upper semicontinuous. We will verify these two conditions but leave out the proof of their sufficiency.

We begin by showing that f_n is an upper semicontinuous function. That is to say, for every r, the set $\{k : f_n(k) < r\}$ is open. Fix a k for which $f_n(k)$ is less than r. Let $S = [-1,1] \setminus \{s \in [-1,1] : h_s(k) \in B_{1/n}\}$. By construction, the measure of S is greater than $2 - r$. Let C be a compact subset of S with measure $2 - r$. Let B' be the subset of pairs $(k', s) \in K \times [-1,1]$ for which $h_s(k')$ is not in $B_{1/n}$. Using the compactness of C and the fact that B' is relatively open we can find a neighborhood U of k such that $C \times U$ is contained in B'. It follows that for $k \in U$, $f_n(k) < r$. This shows semicontinuity.

Fix k and let $T_n = \{s \in [-1,1] : h_s(k) \in B_{1/n}\}$, $f_n = \mu(T_n)$. Now T_n is a decreasing sequence of sets so tht $f_n(k)$ is non-increasing and $\lim f_n(k)$ is the measure of $\bigcap T_n$. Since B is closed, the intersection of the sets T_n is just the set of $s \in [-1,1]$ such that $h_s(k) \in B$. The set of such s is countable and therefore has measure zero. □

We continue with the proof of Theorem 4.5. Choose ϵ so that the fraction of points in each horocycle orbit segment which lies in B_ϵ is less that $1/2$. Let K' be the union of sets $h_s(K)$ for $s \in [-1,1]$. Let K'' be the union of closed unit balls with centers in K'. K'' is compact. Let V_n be the set of θ's in V with the property that for $t > n$ if $g_t(q) \in K''$ then $g_t(q) \in B_{\epsilon/2}$. The union of the sets V_n is V; thus, there is some set V_n with positive measure. Choose one such set and call it V'. Let θ' be a point of density in V'. By changing coordinates we may assume that $\theta' = 0$.

The limit formula (4.2) converges uniformly for s in any compact set. Choose n large enough so that for $t > N$ and $k \in K$ the distance between the left hand side of (4.2) and the right hand side is less than $\epsilon/2$ when applied to any $k \in K$ and form $s \in [-1,1]$. Consider a $t > N$ for which $g_t(q) \in K$. Let $\theta \in [-e^{-t}, e^{-t}]$. Let $s = e^t \cdot \theta$. Applying the left and right sides of (4.2) to $g_t(q)$ shows that the distance between $g_t r_\theta(q)$ and $h_s g_t(q)$ is less than $\epsilon/2$. Claim: $g_t r_\theta(q) \in K''$. By assumption $g_t(q) \in K$. Since $s \in [-1,1]$, $h_s g_t(q)$ is in K'. The distance between $h_s g_t(q)$ and $g_t r_\theta(q)$ is less than one so that $g_t r_\theta(q)$ is in K''. For $\theta \in V'$, $g_t r_\theta(q)$ is in $B_{\epsilon/2}$ when it is in K''. If $g_t r_\theta(q)$ is in $B_{\epsilon/2}$ then $h_s g_t(q)$ is in B_ϵ. This can be the case for at most half of the $s \in [-1,1]$. This shows that the density of V' in the interval $[-e^{-t}, e^{-t}]$ is less than $1/2$. This argument holds for a sequence of arbitrarily large values of t. This contradicts the assumption that θ' was a point of density for V'. □

4.2.2 Geometry of the q-metric

Definition 4.7. Let $Q = Q(k, \chi)$ be the space of quadratic differentials with area one on a surface M of Euler characteristic χ with k singularities.

Proposition 4.8. *The set Q_ϵ consisting of quadratic differentials with no geodesic segment of length less than ϵ is compact.*

Proof. Q_ϵ is clearly closed in Q, for if $q_n \to q_0$ and q_0 has a trajectory β of length less than ϵ, then the length of β is less than ϵ for n large enough. It remains to be shown that Q_ϵ lies over a compact set in $\mathcal{M}_g$. If not, there is a sequence q_n in Q_ϵ lying on Riemann surfaces X_n going to infinity in $\mathcal{M}_g$. Passing to subsequences we may assume X_n converges to a Riemann surface X_∞ with nodes acquired by pinching along a set of disjoint curves $\alpha_1, \ldots, \alpha_p$ (see [Ber81]). Again passing to subsequences, the q_n can be assumed to converge uniformly on compact sets to an integrable q_∞ on X_∞. This means q_∞ has at most simple poles at the punctures. For any δ there are curves homotopic to α_i with q_n length less than δ for large n. This contradiction completes the proof. □

Proposition 4.8 shows that if $r_\theta(q)$ is divergent then, as $t \to \infty$, the flat metric corresponding to $g_t r_\theta(q)$ possess arbitrarily short curves. In the remainder of this section we will show that we can choose short curves which are not crossed by other short curves.

Definition 4.9. A *subcomplex* is a triangulation of a subset of M so that the vertices are singularities, the edges are geodesic segments and the faces are triangles which do not contain singularities. We do not require the vertices of a triangle to be distinct.

We assume that if three sides of the complex bound a triangular region containing no singularities, then that region is included in the complex.

Lemma 4.10. *The maximum number of segments in a subcomplex is* $3(k-\chi)$. *The maximum number of triangles is* $2(k-\chi)$.

Proof. By adding a finite collection of segments to a subcomplex we may assume that the complementary regions are disks. These regions can be further subdivided into triangles. Thus, we may assume that we have a decomposition of M into triangular regions with k vertices. Let e be the number of edges and let t be the number of triangles. Euler's formula gives $k - e + t = \chi$. Each edge is contained in 2 triangles and each triangle has 3 edges; thus, $3t = 2e$. These two equalities imply that $t = 2(k-\chi)$ and $e = 3(k-\chi)$. □

Definition 4.11. An *ϵ-subcomplex* is a subcomplex in which each side has length less than or equal to ϵ.

Definition 4.12. A *boundary edge* is an edge which bounds less than two triangles.

Lemma 4.13. *If* $\epsilon < \left((k-\chi)3^{1/2}/2\right)^{-1/2}$, *then an ϵ-subcomplex has a boundary edge.*

Proof. A triangle with sides of length less than ϵ has area less than $\epsilon^2 3^{1/2}/4$. The number of triangles is less than $2(k-\chi)$. If ϵ is smaller than the number above, then the area of the ϵ-subcomplex is less than one. If the ϵ-subcomplex is not a triangulation of the entire surface then it must have some boundary edge. □

Definition 4.14. A subset X of M is *convex* if every path (not necessarily joining singularities) lying in X, which is homotopic relative to endpoints to a geodesic path, is homotopic to a geodesic path lying in X.

Let B be a vertex in an ϵ-complex. The segments that contain B are cyclically ordered. Let AB and BC be adjacent edges. We say that ABC is an external angle if the segments AB and BC are not contained in a triangle ABC which is contained in the ϵ-complex.

Lemma 4.15. *An ϵ-subcomplex is convex if and only if the measure of each external angle is greater than π.*

Proof. Denote the ϵ-complex by X. The second condition is the criterion for local convexity. It is clear that if X is convex then it is locally convex. To prove the converse choose a path in X homotopic rel endpoints to a geodesic segment not lying in X. Find the shortest path in this relative homotopy class that does lie in X. This path is a piecewise geodesic but not a geodesic. Where this path crosses a singularity we can measure the angle of bending on the right side of the path or on the left side of the path. The sum of these two angles will be the cone angle at the singularity. Since this path can be shortened, the measure of some bending angle ABC at some vertex B is less that π. The path can be shortened by moving it into the interior of the angle ABC. Since this shortened path is not contained in X there must be some sector of the angle ABC not contained in X. □

Lemma 4.16. *If a boundary edge of a connected ϵ-subcomplex is crossed by a geodesic segment of length less than C, where $C > \epsilon$, then there is a connected $(C + \epsilon)$-subcomplex with a larger number of simplices.*

Proof. Let X be the ϵ-subcomplex. If X is not convex, then we can find some boundary angle ABC with measure less than π. If there is a segment AC so that the region bounded by the segments AB, BC and AC is a disk with no singularities then add the segment AC and the triangle ABC. The length of AC is less than 2ϵ. Now assume that this is not the case. Let M_r be the point on the segment AB such that the ratio of the length of BM_r to the length of AB is r. Let N_r be the point on the segment CB such that the ratio of the length of BN_r to the length of CB is r. For small values of r we can construct a triangle M_rBN_r which contains no singularities. Let r_0 be the first r for which such a triangle cannot be constructed. The obstruction is a singularity P in the closure of the union of segments M_rN_r for $r < r_0$. We can construct a segment BP. The segment BP cannot intersect any segment in the ϵ- complex. Add BP to the ϵ-complex. The length of BP is less than ϵ.

Assume that X is convex. If a geodesic segment crosses the boundary of X then at least one endpoint is not contained in X. Thus, there is a singularity not in X withing the distance C of X. Consider the singularity closest to X. Let this point be P. Let AB be the segment in the boundary of X which contains the point closest to P. The triangle ABP cannot contain any singularities not contained in X; otherwise, these would be closer to X than P. The triangle cannot contain any point Q in X for if it did, by convexity it would contain the triangle ABQ. This would contradict the assumption that the closest point to P lies on AB. We can add the triangle ABP to the original subcomplex. The longest side of ABP is less than $C + \epsilon$. □

Definition 4.17. Let ϵ be less than C. A geodesic segement AB is (ϵ, C)-*isolated* if it has length less than ϵ and every geodesic that crosses AB in its interior has length greater than C.

Definition 4.18. $N(\epsilon, C)$ is the set of quadratic differentials which possess (ϵ, C)-isolated curves.

We consider (ϵ, C)-isolated segments because there is a fixed bound to the number that can exist in any of the flat metrics. This follows from the fact that distinct (ϵ, C)-isolated segments are disjoint and the number of disjoint segments is less than or equal to $3(k - \chi)$. This bound plays a key role in the proof of Theorem 4.22.

Proposition 4.19. *Suppose that there is a set S of positive measure so that $r_\theta(q)$ is divergent for $\theta \in S$. Then, there are:*

1. *a sequence of times T_i increasing to infinity,*
2. *a sequence of subsets S_i contained in S such that $\mu(S_i) \geq \delta > 0$,*
3. *a sequence of $\epsilon_i > 0$ converging to 0,*
4. *a positive constant C such that:*

$$g_{T_i} r_\theta(q) \in N(\epsilon_i, C) \text{ for } \theta \in S_i.$$

The idea behind Proposition 4.19 is that if $r_\theta(q)$ is divergent, then there will be times T_i after which there will always be a segment of length less than ϵ_i in the corresponding metrics. Unfortunately, for a fixed C, the sets $N(\epsilon_i, C)$ do not form a neighborhood basis for infinity; so we cannot make the same statement for (ϵ_i, C)-isolated segments. However, by considering subsequences of times and subsets of S, we are able to make an analogous statement.

Definition 4.20. Let $U(\epsilon)$ be the set of quadratic differentials with a geodesic segment of length less than ϵ.

Definition 4.21. Let $n_\epsilon(q)$ be the maximum number of simplices in a connected ϵ-subcomplex.

Proof of Proposition 4.19. Choose a sequence $\epsilon_i \to 0$. By Proposition 4.8, the sets $U(\epsilon_i)$ have compact complements. We conclude that for each divergent ray $r_\theta(q)$ there are times T_i such that $g_t r_\theta(q) \in U(\epsilon_i)$, for all $t > T_i$. The T_i can be chosen so that this relationship holds for all $\theta \in S'$, where $S' \subset S$ and $\mu(S') > \mu(S)/2$.

We have now constructed:

1. a sequence of times T_i increasing to infinity,
2. a set S' of θ's with positive measure,

3. a sequence of positive numbers ϵ_i converging to zero such that

$$g_{T_i} r_\theta(q) \in U(\epsilon_i) \text{ for } \theta \in S_i'.$$

In order to prove Proposition 4.19 we will construct sets $S_i \subset S'$ and find a constant C such that:

$$g_{T_i} r_\theta(q) \in N(\epsilon_i, C) \text{ for } \theta \in S_i.$$

Consider triples of sequences ϵ_i, T_i and S_i with the property that ϵ_i goes to zero, T_i goes to infinity, the measure of S_i is bounded below and for $\theta \in S_i$ the surface $g_{T_i} r_\theta(q)$ is an element of $U(\epsilon_i)$. We can construct one such triple by taking ϵ_i and T_i as above and taking $S_i = S$ for all i. For each such triple we can compute the minimum $n_\epsilon(g_{T_i} r_\theta(q))$. Choose a triple of sequences that maximizes this number. For each i and each θ in S_i, let $C_{i,\theta}$ be the length of the shortest geodesic segment that crosses a boundary segment of the ϵ_i-subcomplex of $g_{T_i} r_\theta(q)$.

For each i, let m_i be the median value of $C_{i,\theta}$ on S_i. That is, m_i is some value so that the measure of points with greater or equal value is equal to the measure of points with smaller or equal value. We claim that $m_i \geq C$ for some $C > 0$. Suppose not. Then we could replace S_i by the set of θ's with values of $C_{i,\theta}$ less than or equal to m_i and replace ϵ_i by $\epsilon_i + m_i$. We could then choose a subsequence so that m_i converged to zero. The new sequence of ϵ_i's still goes to zero and the measure of the new sets S_i is still bounded below. According to Lemma 4.16, we could find ϵ-subcomplexes with more simplices. This contradicts the maximality of our triple of sequences. Thus, the set of m_i is bounded below as claimed.

If we replace S_i by the set of θ with $C_{i,\theta} \geq m \geq C$ we ensure that every geodesic segment crossing the ϵ_i-complex has length at least C. □

4.2.3 Case II: r_θ divergent

In this section we consider the case of divergent $r_\theta(q)$ and prove:

Theorem 4.22. *The set of θ's such that $r_\theta(q)$ is divergent has measure zero.*

Proof. The argument is by contradiction. We assume that there is a set of θ's of positive measure for which $r_\theta(q)$ is divergent. With this assumption Proposition 4.19 applies.

Proposition 4.23. *Fix $C > 0$. There are constants T, ϵ_1 and K such that, for $t > T$ and $\epsilon < \epsilon_1$, the measure of the set of θ such that $g_t r_\theta(q)$ is in $N(\epsilon, C)$ is less than $K\epsilon$.*

When $K\epsilon$ is less than the lower bound ρ for the measures of the sets S_i given by Proposition 4.19, we have a contradiction. Thus, no non-trivial set of divergent geodesics can exist. □

The remainder of this section is devoted to the proof of Proposition 4.23.

Behavior of the length function: Let α be a geodesic segment. Denote by $l(t, \theta)$ its length with respect to the structure $g_t r_\theta(q)$. The segment α is vertical in $r_\theta(q)$; i.e. $h = 0$

and $v = l(0,\theta)$, for some value $\theta(\alpha)$. Rotate the circular coordinate so that $\theta(\alpha) = 0$. In these coordinates

$$l(t,\theta) = v \cdot (e^t \sin^2\theta + e^{-t}\cos^2\theta)^{1/2}. \tag{4.3}$$

Fix a value t and choose $\epsilon > 0$. For each geodesic segment α, define two intervals $I_\alpha \subset J_\alpha \subset (-\pi/2, \pi/2)$ with respect to the rotated coordinates for α.

$$I_\alpha = \{\theta : |\sin\theta| \le \epsilon/(ve^{t/2})\},$$
$$J_\alpha = \{\theta : |\sin\theta| \le C/(2ve^{t/2})\}.$$

Lemma 4.24. *There are constants ϵ_1, T and K, independent of α, such that for $\epsilon < \epsilon_1$ and $t > T$ we have $\mu(I_\alpha)/\mu(J_\alpha) < K\epsilon$. If $l(t,\theta) < \epsilon$ for some θ, then it is less than C on J_α and is greater than ϵ outside I_α*

Proof. If $l(t,\theta) < \epsilon$ for some θ, then

$$ve^{-t/2}|\cos\theta| \le ve^{-t/2} = l(t,\theta) < \epsilon$$

since $l(t,\theta)$ attains its minimum at 0. From (4.3) we have

$$l(t,\theta) < C/2 + \epsilon \quad \text{for } \theta \in J_\alpha.$$

If we choose $\epsilon_1 = C/2$ and $\epsilon < \epsilon_1$, then $l(t,\theta)$ is less than C on J_α as desired.

Let $I'_\alpha = \sin I_\alpha$ and $J'_\alpha = \sin J_\alpha$. If $C/2ve^{t/2} < 1/2$ then J'_α is contained in the interval $(-1/2, 1/2)$ and $\mu(I'_\alpha)/\mu(J'_\alpha) < 2\epsilon/C$. Since the function arcsin restricted to the interval $(-1/2, 1/2)$ is Lipschitz and has a Lipschitz inverse, it changes the lengths of intervals by a bounded amount. Thus, we can choose a K depending on C such that $\mu(I_\alpha)/\mu(J_\alpha) < K\epsilon$, as claimed. To ensure that $C/2ve^{t/2} < 1/2$, choose T such that $C/2me^{T/2} < 1/2$, where m is the length of the shortest geodesic segment on q. Since $v \ge m$ and $t > T$, the inequality holds. □

Proof of Proposition 4.23. Fix $t > T$ and $\epsilon < \epsilon_1$ so that the above lemma applies. Consider all geodesic segmentes α which are (ϵ, C)-isolated with respect to $g_t r_\theta(q)$ for some θ. For each α let $\hat{I}_\alpha$ be the smallest interval of θ's which contains the set of θ's for which α is (ϵ, C)-isolated. Then $\hat{I}\alpha \subset I_\alpha \subset J_\alpha$, where I_α and J_α are defined as above.

Construct a new open interval $\hat{J}_\alpha$ as follows. The left-hand endpoint of $\hat{J}_\alpha$ will be halfway between the left-hand endpoints of $\hat{I}_\alpha$ and J_α. The point of this construction is the following fact. If $\hat{J}_\alpha$ and $\hat{J}_\beta$ intersect, then either J_α intersects $\hat{I}_\beta$ or J_β intersects $\hat{I}_\alpha$. Assume that J_α intersects $\hat{I}_\beta$. We can find a θ in the intersection so that β is (ϵ, C)-isolated with respect to $g_t r_\theta(q)$. Now with respect to $g_t r_\theta(q)$, α has length less than C and β cannot cross any geodesic segment of length less than C. Therefore, the interiors of α and β are disjoint. By Lemma 4.10 the maximum number of disjoint geodesic segments is $G = 3(k - \chi)$.

We conclude that no θ can lie in more than G sets $\hat{J}_\alpha$. The sum of the lengths of the intervals $\hat{J}_\alpha$ is at most πG. The sum of the lengths of the intervals J_α is at most $2\pi G$. Thus, by Lemma 4.24, the sums of the lengths of the intervals I_α is less than $\epsilon K 2\pi G$. The intervals I_α cover the set of θ's for which $g_t r_\theta(q)$ is in $N(\epsilon, C)$. □

Conclusions and further work

There are not many conclusions and interpretations to be made here, since the results speak by themselves. However, there is still a point to be addressed. In the introduction we said that with the content of this thesis we would be able to answer Q3 and Q4 for rational billiards. Indeed, Q3 has been explicitly answered by Theorem 2.18 but we cannot say the same about Q4. The question of equidistribution is a technical one, but roughly speaking the billiard flow is equidistributed if for each region in $\mathcal{P}$ the time spent by the trajectories on that region is proportional to its area. Morally this means that the flow does not have preference for any particular region nor it gets trapped for an infinite amount of time.

Luckily, unique ergodicity implies equidistribution, so Theorem 4.1 answers Q4. Nevertheless, this implication is non-trivial and deserves its explanation, thus we will consider it for further work. Many other interesting phenomena have been left out as well, either for the sake of space or sincerely, due to a lack of knowledge. For instance, the construction of a minimal non-uniquely ergodic billiard or the existence of a non-periodic trajectory that is not dense in the configuration space (which answers Q2 in the negative). Indeed, billiards for which this dichotomy occurs are very special and arise from certain lattices of $\mathrm{PSL}(2, \mathbb{R})$ (see [MT02, §5]).

In §2.1.2 we studied the topology of the billiard class. From this section, we would like to keep working to prove that under conditions A1 and A2 we have that $\partial\mathcal{P}$ is homotopy equivalent to a wedge of circles. Prof. Raúl Oset pointed out that this comes down to noticing that under these special constraints $\partial\mathcal{P}$ is a finite CW-complex of dimension 1 and using theorems from Milnor and Whitehead to stablish first an isomorphism of fundamental groups and after proving they have the same homotopy type.

Another noticeable omission, this for lack of space and time, is that of periodic orbits. These are clearly the most distinguished set of orbits in a billiard and most of the question in the Introduction refer to them. We want to study in the future the result of Masur [Mas86] about the existence of such orbits. Another remarkable feature was already described (in an abstract setting) in §1.2.1: rational billiards possess a dense set of periodic orbits and a dense set of minimal directions (see [Bos+98]).

Although billiards are interesting objects by themselves, nowadays the lines of research have gone in the direction of understanding more general objects that arise from them, namely, translation structures and their moduli spaces. These moduli spaces can be stratified by singularity type and the topology of the strata is still mysterious, although with partial results (see [KZ03]).

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